Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Live Oak Dairy F

Physical address of dairy:

Number and Street	City	County	Zip Code
13310 W Eagle Field RD	Firebaugh	Merced	93622

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1973

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X090-X140-X024-XXXX X090-X140-X025-XXXX X090-X140-X026-XXXX X090-X360-X003-XXXX

B. OPERATORS

Nyman, Brad

Operator name: Nyman, Brad

Telephone no.: Landline: ; Cellular: (209) 495-8821

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 1331	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Nyman, Brad

Legal owner name: Nyman, Brad

Telephone no.: Landline: ; Cellular: (209) 495-8821

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 1331	Hilmar	CA	95324

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	793	708	253	335
Number under roof	1,800	190	0	0	0	0
Maximum number	1,800	198	793	708	253	335
Average number	1,800	198	793	708	253	335
Avg live weight (lbs)	1,350	1,350	1,000	650		

Predominant milk cow breed: Holstein

Average milk production: 71 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 66,114.28 *tons per reporting period*

Total nitrogen from manure: 815,072.91 *lbs per reporting period*

After ammonia losses (30% loss applied): 570,551.04 *lbs per reporting period*

Total phosphorus from manure: 133,708.58 *lbs per reporting period*

Total potassium from manure: 345,899.21 *lbs per reporting period*

Total salt from manure: 893,060.10 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 113,310,000 *gallons*

Total nitrogen generated: 208,996.94 *lbs*

Total phosphorus generated: 40,677.02 *lbs*

Total potassium generated: 453,311.25 *lbs*

Total salt generated: 4,689,901.68 *lbs*

	113,310,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	113,310,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW #1	Ground water
IW #2	Ground water
IW #3	Ground water
IW #4	Ground water
Ponoche WD	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
350	80	80	2	manure	X090-X140-X045-XXXX
351	80	80	1	manure	X090-X140-X045-XXXX
352	73	73	1	process wastewater	X090-X140-X030-XXXX
353	66	66	1	process wastewater	X090-X140-X030-XXXX
354	61	61	2	process wastewater	X090-X140-X030-XXXX
355	61	61	2	process wastewater	X090-X140-X027-XXXX X090-X140-X030-XXXX
356	61	61	1	process wastewater	X090-X140-X027-XXXX
357	59	59	1	process wastewater	X090-X140-X018-XXXX
358	76	76	1	process wastewater	X090-X140-X025-XXXX
359	73	73	1	process wastewater	X090-X140-X024-XXXX
360	66	66	2	manure	X090-X140-X026-XXXX
361	68	68	2	manure	X090-X200-X026-XXXX X090-X200-X027-XXXX
362	62	62	1	process wastewater	X090-X140-X028-XXXX
363	52	52	1	process wastewater	X090-X140-X028-XXXX
364	97	97	1	process wastewater	X090-X140-X028-XXXX X090-X140-X029-XXXX
365	92	92	1	process wastewater	X090-X140-X029-XXXX
366	61	61	1	process wastewater	X090-X140-X019-XXXX
367	65	65	2	process wastewater	X090-X140-X025-XXXX
368	70	70	2	process wastewater	X090-X140-X024-XXXX X090-X140-X026-XXXX
369	65	65	1	manure	X004-X020-X015-XXXX X090-X200-X025-XXXX
Totals for areas that were used for application	1,388	1,388	27		
Totals for areas that were not used for application					
Land application area totals	1,388	1,388	27		

B. CROPS AND HARVESTS

350

Field name: 350

350

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 80 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	980.00 *ton*	As-is		58.3	5,900.00	1,200.00	7,500.00		7.39

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	12.25	144.55	29.40	183.75	755.00

07/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 07/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	1,989.00 *ton*	As-is		70.0	3,330.00	720.00	3,670.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	24.86	165.58	35.80	182.49	835.38

351

Field name: 351

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 80 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	980.00 *ton*	As-is		58.3	5,930.00	1,220.00	7,480.00		7.39

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	12.25	145.29	29.89	183.26	755.00

352

Field name: 352

11/01/2017: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 73 Plant date: 11/01/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	374.00 *ton*	As-is		15.0	26,400.00	2,550.00	17,000.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.50	450.00	40.50	315.00	0.00
Total actual harvest content	5.12	270.51	26.13	174.19	958.05

353

Field name: 353

11/01/2017: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 66 Plant date: 11/01/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	387.00 *ton*	As-is		13.7	33,400.00	2,680.00	16,400.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.50	450.00	40.50	315.00	0.00
Total actual harvest content	5.86	391.69	31.43	192.33	1,133.51

354

Field name: 354

354

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 61 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/23/2023	780.00 *ton*	As-is		64.4	5,790.00	1,220.00	8,120.00		8.79

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	12.79	148.07	31.20	207.66	800.26

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 61 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,303.00 *ton*	As-is		70.0	3,150.00	670.00	3,600.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	21.36	134.57	28.62	153.80	589.55

355

Field name: 355

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 61 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/23/2023	780.00 *ton*	As-is		60.3	7,590.00	1,720.00	11,900.00		13.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	12.79	194.10	43.99	304.33	1,340.17

355

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 61 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,303.00 *ton*	As-is		70.0	2,940.00	730.00	3,570.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	21.36	125.60	31.19	152.52	653.64

356

Field name: 356

12/01/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 61 Plant date: 12/01/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	248.00 *ton*	As-is		14.0	36,500.00	3,010.00	17,200.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.50	450.00	40.50	315.00	0.00
Total actual harvest content	4.07	296.79	24.47	139.86	776.20

357

Field name: 357

357

12/01/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 59 Plant date: 12/01/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	266.00 *ton*	As-is		16.1	34,400.00	2,850.00	16,800.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.50	450.00	40.50	315.00	0.00
Total actual harvest content	4.51	310.18	25.70	151.48	915.39

358

Field name: 358

12/01/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 76 Plant date: 12/01/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	321.00 *ton*	As-is		12.8	28,500.00	1,830.00	26,200.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.50	450.00	40.50	315.00	0.00
Total actual harvest content	4.22	240.75	15.46	221.32	913.40

359

Field name: 359

359

12/01/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 73 Plant date: 12/01/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	319.00 *ton*	As-is		13.8	28,600.00	1,550.00	22,400.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.50	450.00	40.50	315.00	0.00
Total actual harvest content	4.37	249.96	13.55	195.77	911.57

360

Field name: 360

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 66 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/26/2023	866.00 *ton*	As-is		66.8	5,590.00	1,270.00	7,920.00		9.83

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.12	146.70	33.33	207.84	856.44

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 66 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	1,186.00 *ton*	As-is		70.0	3,330.00	700.00	3,520.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	17.97	119.68	25.16	126.51	743.95

361

Field name: 361

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 68 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/26/2023	866.00 *ton*	As-is		64.4	5,700.00	1,300.00	7,600.00		9.53

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	12.74	145.18	33.11	193.58	864.14

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 68 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	1,186.00 *ton*	As-is		70.0	3,120.00	730.00	3,450.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	17.44	108.83	25.46	120.34	722.06

362

Field name: 362

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 62 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	811.00 *ton*	As-is		66.3	5,380.00	1,260.00	6,900.00		8.88

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	13.08	140.75	32.96	180.51	782.89

363

Field name: 363

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 52 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	625.00 *ton*	As-is		54.7	7,300.00	1,500.00	8,900.00		8.29

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.02	175.48	36.06	213.94	902.73

364

Field name: 364

11/20/2002: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 97 Plant date: 11/20/2002

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	1,223.00 *ton*	As-is		61.7	6,230.00	1,280.00	7,600.00		9.34

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	12.61	157.10	32.28	191.65	902.05

365

Field name: 365

365

12/01/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 92 Plant date: 12/01/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	532.00 *ton*	As-is		23.4	23,100.00	1,460.00	20,700.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.50	450.00	40.50	315.00	0.00
Total actual harvest content	5.78	267.16	16.89	239.40	1,054.22

366

Field name: 366

12/01/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 61 Plant date: 12/01/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	341.00 *ton*	As-is		8.2	29,600.00	1,930.00	31,200.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.50	450.00	40.50	315.00	0.00
Total actual harvest content	5.59	330.94	21.58	348.83	1,221.36

367

Field name: 367

367

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 65 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	1,200.00 *ton*	As-is		67.3	5,800.00	1,060.00	6,210.00		9.49

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.50	75.00	12.00	62.25	0.00
Total actual harvest content	18.46	214.15	39.14	229.29	1,145.81

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 65 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	1,412.00 *ton*	As-is		70.0	3,300.00	650.00	3,150.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	21.72	143.37	28.24	136.86	612.59

368

Field name: 368

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 70 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	1,260.00 *ton*	As-is		61.4	6,390.00	1,300.00	7,470.00		8.96

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.50	75.00	12.00	62.25	0.00
Total actual harvest content	18.00	230.04	46.80	268.92	1,245.08

368

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	1,212.00 *ton*	As-is		70.0	3,960.00	740.00	3,660.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	17.31	137.13	25.63	126.74	623.31

369

Field name: 369

11/20/2002: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 65 Plant date: 11/20/2002

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/26/2023	1,360.00 *ton*	As-is		58.5	6,150.00	1,420.00	8,110.00		9.61

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.92	257.35	59.42	339.37	1,668.89

NUTRIENT BUDGET

A. LAND APPLICATIONS

350 - 11/20/2022: Oats, silage-soft dough

Field name: 350

Crop: Oats, silage-soft dough — Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Fall Solid Manure	Corral solids	201.30	63.96	380.95	3,225.11	650.00 *ton*
Application event totals		201.30	63.96	380.95	3,225.11	

03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,470.56	10,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,470.56	

350 - 07/03/2023: Corn, silage

Field name: 350

Crop: Corn, silage — Plant date: 07/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Spring Solid Manure	Corral solids	129.36	29.11	18.94	1,980.00	1,000.00 *ton*
Application event totals		129.36	29.11	18.94	1,980.00	

06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,784.49	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,784.49	

350 - 07/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,586.22	11,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,586.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,520.13	11,040,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,520.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,718.40	12,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,718.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,784.49	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,784.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,553.17	11,280,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,553.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,553.17	11,280,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,553.17	

350 - 07/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	148.71	1,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	148.71	

351 - 11/20/2022: Oats, silage-soft dough

Field name: 351

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Fall Solid Manure	Corral solids	201.30	63.96	380.95	3,225.11	650.00 *ton*
Application event totals		201.30	63.96	380.95	3,225.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,470.56	10,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,470.56	

352 - 11/01/2017: Alfalfa, hay

Field name: 352

Crop: Alfalfa, hay

Plant date: 11/01/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

352 - 11/01/2017: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	148.77	23.46	238.90	3,740.85	5,400,000.00 *gal*
Application event totals		148.77	23.46	238.90	3,740.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.99	0.00	0.00	1,711.98	11,520,000.00 *gal*
Application event totals		3.99	0.00	0.00	1,711.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.99	0.00	0.00	1,711.98	11,520,000.00 *gal*
Application event totals		3.99	0.00	0.00	1,711.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.49	0.00	0.00	1,497.98	10,080,000.00 *gal*
Application event totals		3.49	0.00	0.00	1,497.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.49	0.00	0.00	1,497.98	10,080,000.00 *gal*
Application event totals		3.49	0.00	0.00	1,497.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	2.99	0.00	0.00	1,283.99	8,640,000.00 *gal*
Application event totals		2.99	0.00	0.00	1,283.99	

352 - 11/01/2017: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.33	0.00	0.00	1,426.65	9,600,000.00 *gal*
Application event totals		3.33	0.00	0.00	1,426.65	

10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.33	0.00	0.00	1,426.65	9,600,000.00 *gal*
Application event totals		3.33	0.00	0.00	1,426.65	

353 - 11/01/2017: Alfalfa, hay

Field name: 353

Crop: Alfalfa, hay

Plant date: 11/01/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	164.55	25.95	264.23	4,137.60	5,400,000.00 *gal*
Application event totals		164.55	25.95	264.23	4,137.60	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	4.41	0.00	0.00	1,893.56	11,520,000.00 *gal*
Application event totals		4.41	0.00	0.00	1,893.56	

04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	4.41	0.00	0.00	1,893.56	11,520,000.00 *gal*
Application event totals		4.41	0.00	0.00	1,893.56	

353 - 11/01/2017: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.86	0.00	0.00	1,656.86	10,080,000.00 *gal*
Application event totals		3.86	0.00	0.00	1,656.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.86	0.00	0.00	1,656.86	10,080,000.00 *gal*
Application event totals		3.86	0.00	0.00	1,656.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.31	0.00	0.00	1,420.17	8,640,000.00 *gal*
Application event totals		3.31	0.00	0.00	1,420.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.68	0.00	0.00	1,577.96	9,600,000.00 *gal*
Application event totals		3.68	0.00	0.00	1,577.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.68	0.00	0.00	1,577.96	9,600,000.00 *gal*
Application event totals		3.68	0.00	0.00	1,577.96	

354 - 11/20/2022: Oats, silage-soft dough

Field name: 354

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

354 - 11/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.31	9.83	100.06	1,566.86	1,890,000.00 *gal*
IW #4	Ground water	2.09	0.00	0.00	896.34	5,040,000.00 *gal*
Application event totals		64.40	9.83	100.06	2,463.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.85	8.16	94.41	1,298.70	1,620,000.00 *gal*
IW #4	Ground water	1.79	0.00	0.00	768.29	4,320,000.00 *gal*
Application event totals		57.64	8.16	94.41	2,066.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.46	10.87	125.88	1,731.60	2,160,000.00 *gal*
IW #4	Ground water	2.39	0.00	0.00	1,024.38	5,760,000.00 *gal*
Application event totals		76.85	10.87	125.88	2,755.98	

354 - 06/23/2023: Corn, silage

Field name: 354

Crop: Corn, silage

Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	94.19	25.78	384.14	1,898.56	2,700,000.00 *gal*
IW #4	Ground water	2.98	0.00	0.00	1,280.48	7,200,000.00 *gal*
Application event totals		97.17	25.78	384.14	3,179.03	

354 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.58	0.00	0.00	1,536.57	8,640,000.00 *gal*
Application event totals		3.58	0.00	0.00	1,536.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.28	0.00	0.00	1,408.53	7,920,000.00 *gal*
Application event totals		3.28	0.00	0.00	1,408.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.72	12.90	73.47	339.08	2,430,000.00 *gal*
IW #4	Ground water	2.69	0.00	0.00	1,152.43	6,480,000.00 *gal*
Application event totals		33.40	12.90	73.47	1,491.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	2.89	0.00	0.00	1,237.80	6,960,000.00 *gal*
Application event totals		2.89	0.00	0.00	1,237.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.58	12.42	70.75	326.52	2,340,000.00 *gal*
IW #4	Ground water	2.59	0.00	0.00	1,109.75	6,240,000.00 *gal*
Application event totals		32.17	12.42	70.75	1,436.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	2.69	0.00	0.00	1,152.43	6,480,000.00 *gal*
Application event totals		2.69	0.00	0.00	1,152.43	

355 - 11/20/2022: Oats, silage-soft dough

Field name: 355

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.31	9.83	100.06	1,566.86	1,890,000.00 *gal*
IW #4	Ground water	2.09	0.00	0.00	896.34	5,040,000.00 *gal*
Application event totals		64.40	9.83	100.06	2,463.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.85	8.16	94.41	1,298.70	1,620,000.00 *gal*
IW #4	Ground water	1.79	0.00	0.00	768.29	4,320,000.00 *gal*
Application event totals		57.64	8.16	94.41	2,066.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.46	10.87	125.88	1,731.60	2,160,000.00 *gal*
IW #4	Ground water	2.39	0.00	0.00	1,024.38	5,760,000.00 *gal*
Application event totals		76.85	10.87	125.88	2,755.98	

355 - 06/23/2023: Corn, silage

Field name: 355

Crop: Corn, silage

Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	94.19	25.78	384.14	1,898.56	2,700,000.00 *gal*
IW #4	Ground water	2.98	0.00	0.00	1,280.48	7,200,000.00 *gal*
Application event totals		97.17	25.78	384.14	3,179.03	

355 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.58	0.00	0.00	1,536.57	8,640,000.00 *gal*
Application event totals		3.58	0.00	0.00	1,536.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	3.28	0.00	0.00	1,408.53	7,920,000.00 *gal*
Application event totals		3.28	0.00	0.00	1,408.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.72	12.90	73.47	339.08	2,430,000.00 *gal*
IW #4	Ground water	2.69	0.00	0.00	1,152.43	6,480,000.00 *gal*
Application event totals		33.40	12.90	73.47	1,491.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	2.89	0.00	0.00	1,237.80	6,960,000.00 *gal*
Application event totals		2.89	0.00	0.00	1,237.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.58	12.42	70.75	326.52	2,340,000.00 *gal*
IW #4	Ground water	2.59	0.00	0.00	1,109.75	6,240,000.00 *gal*
Application event totals		32.17	12.42	70.75	1,436.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4	Ground water	2.69	0.00	0.00	1,152.43	6,480,000.00 *gal*
Application event totals		2.69	0.00	0.00	1,152.43	

356 - 12/01/2020: Alfalfa, hay

Field name: 356

Crop: Alfalfa, hay

Plant date: 12/01/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	178.04	28.07	285.89	4,476.75	5,400,000.00 *gal*
Application event totals		178.04	28.07	285.89	4,476.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,323.82	8,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,323.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,323.82	8,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,323.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,544.45	10,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,544.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,287.05	8,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,287.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,250.27	8,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,250.27	

356 - 12/01/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,176.73	7,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,176.73	

09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,323.82	8,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,323.82	

357 - 12/01/2020: Alfalfa, hay

Field name: 357

Crop: Alfalfa, hay

Plant date: 12/01/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	184.07	29.02	295.58	4,628.50	5,400,000.00 *gal*
Application event totals		184.07	29.02	295.58	4,628.50	

03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,368.69	8,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,368.69	

04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,368.69	8,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,368.69	

357 - 12/01/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,596.81	10,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,596.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,330.67	8,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,330.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,292.65	8,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,292.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,216.62	7,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,216.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #3	Ground water	0.00	0.00	0.00	1,368.69	8,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,368.69	

358 - 12/01/2019: Alfalfa, hay

Field name: 358

Crop: Alfalfa, hay

Plant date: 12/01/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

358 - 12/01/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.09	0.00	0.00	1,535.83	11,280,000.00 *gal*
Application event totals		2.09	0.00	0.00	1,535.83	

05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.14	0.00	0.00	1,568.51	11,520,000.00 *gal*
Application event totals		2.14	0.00	0.00	1,568.51	

06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.94	0.00	0.00	1,421.46	10,440,000.00 *gal*
Application event totals		1.94	0.00	0.00	1,421.46	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.09	0.00	0.00	1,535.83	11,280,000.00 *gal*
Application event totals		2.09	0.00	0.00	1,535.83	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.86	0.00	0.00	1,364.28	10,020,000.00 *gal*
Application event totals		1.86	0.00	0.00	1,364.28	

09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.60	0.00	0.00	1,176.38	8,640,000.00 *gal*
Application event totals		1.60	0.00	0.00	1,176.38	

358 - 12/01/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.72	24.83	369.99	1,828.61	3,240,000.00 *gal*
Application event totals		90.72	24.83	369.99	1,828.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.60	0.00	0.00	1,176.38	8,640,000.00 *gal*
Application event totals		1.60	0.00	0.00	1,176.38	

359 - 12/01/2019: Alfalfa, hay

Field name: 359

Crop: Alfalfa, hay

Plant date: 12/01/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.18	0.00	0.00	1,598.95	11,280,000.00 *gal*
Application event totals		2.18	0.00	0.00	1,598.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.23	0.00	0.00	1,632.97	11,520,000.00 *gal*
Application event totals		2.23	0.00	0.00	1,632.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.02	0.00	0.00	1,479.88	10,440,000.00 *gal*
Application event totals		2.02	0.00	0.00	1,479.88	

359 - 12/01/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.18	0.00	0.00	1,598.95	11,280,000.00 *gal*
Application event totals		2.18	0.00	0.00	1,598.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.94	0.00	0.00	1,420.34	10,020,000.00 *gal*
Application event totals		1.94	0.00	0.00	1,420.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.67	0.00	0.00	1,224.73	8,640,000.00 *gal*
Application event totals		1.67	0.00	0.00	1,224.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	94.45	25.85	385.20	1,903.76	3,240,000.00 *gal*
Application event totals		94.45	25.85	385.20	1,903.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.67	0.00	0.00	1,224.73	8,640,000.00 *gal*
Application event totals		1.67	0.00	0.00	1,224.73	

360 - 11/20/2022: Oats, silage-soft dough

Field name: 360

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

360 - 11/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Fall Solid Manure	Corral solids	206.46	65.60	390.72	3,307.80	550.00 *ton*
Application event totals		206.46	65.60	390.72	3,307.80	

03/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.85	0.00	0.00	1,354.62	8,640,000.00 *gal*
Application event totals		1.85	0.00	0.00	1,354.62	

360 - 06/22/2023: Corn, silage

Field name: 360

Crop: Corn, silage

Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Spring Solid Manure	Corral solids	133.28	29.99	19.52	2,040.00	850.00 *ton*
Application event totals		133.28	29.99	19.52	2,040.00	

06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.57	0.00	0.00	152.94	10,080,000.00 *gal*
Application event totals		3.57	0.00	0.00	152.94	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.27	0.00	0.00	140.20	9,240,000.00 *gal*
Application event totals		3.27	0.00	0.00	140.20	

360 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.14	0.00	0.00	134.73	8,880,000.00 *gal*
Application event totals		3.14	0.00	0.00	134.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.06	0.00	0.00	131.09	8,640,000.00 *gal*
Application event totals		3.06	0.00	0.00	131.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.31	0.00	0.00	142.02	9,360,000.00 *gal*
Application event totals		3.31	0.00	0.00	142.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.06	0.00	0.00	131.09	8,640,000.00 *gal*
Application event totals		3.06	0.00	0.00	131.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.06	0.00	0.00	131.09	8,640,000.00 *gal*
Application event totals		3.06	0.00	0.00	131.09	

361 - 11/20/2022: Oats, silage-soft dough

Field name: 361

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

361 - 11/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Fall Solid Manure	Corral solids	200.39	63.67	379.23	3,210.51	550.00 *ton*
Application event totals		200.39	63.67	379.23	3,210.51	

03/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.09	0.00	0.00	1,533.91	10,080,000.00 *gal*
Application event totals		2.09	0.00	0.00	1,533.91	

361 - 06/22/2023: Corn, silage

Field name: 361

Crop: Corn, silage

Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Spring Solid Manure	Corral solids	129.36	29.11	18.94	1,980.00	850.00 *ton*
Application event totals		129.36	29.11	18.94	1,980.00	

06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.46	0.00	0.00	148.44	10,080,000.00 *gal*
Application event totals		3.46	0.00	0.00	148.44	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.18	0.00	0.00	136.07	9,240,000.00 *gal*
Application event totals		3.18	0.00	0.00	136.07	

361 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.05	0.00	0.00	130.77	8,880,000.00 *gal*
Application event totals		3.05	0.00	0.00	130.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	2.97	0.00	0.00	127.24	8,640,000.00 *gal*
Application event totals		2.97	0.00	0.00	127.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.22	0.00	0.00	137.84	9,360,000.00 *gal*
Application event totals		3.22	0.00	0.00	137.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	2.97	0.00	0.00	127.24	8,640,000.00 *gal*
Application event totals		2.97	0.00	0.00	127.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	2.97	0.00	0.00	127.24	8,640,000.00 *gal*
Application event totals		2.97	0.00	0.00	127.24	

362 - 11/20/2022: Oats, silage-soft dough

Field name: 362

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

362 - 11/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.15	10.59	107.82	1,688.41	2,070,000.00 *gal*
IW #3	Ground water	0.00	0.00	0.00	832.13	5,520,000.00 *gal*
Application event totals		67.15	10.59	107.82	2,520.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.47	8.25	95.47	1,313.25	1,665,000.00 *gal*
IW #3	Ground water	0.00	0.00	0.00	669.32	4,440,000.00 *gal*
Application event totals		56.47	8.25	95.47	1,982.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.63	9.58	110.95	1,526.21	1,935,000.00 *gal*
IW #3	Ground water	0.00	0.00	0.00	777.86	5,160,000.00 *gal*
Application event totals		65.63	9.58	110.95	2,304.07	

363 - 11/20/2022: Oats, silage-soft dough

Field name: 363

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	73.10	11.53	117.38	1,838.05	1,890,000.00 *gal*
IW #3	Ground water	0.00	0.00	0.00	905.88	5,040,000.00 *gal*
Application event totals		73.10	11.53	117.38	2,743.93	

363 - 11/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.51	9.57	110.75	1,523.48	1,620,000.00 *gal*
IW #3	Ground water	0.00	0.00	0.00	776.47	4,320,000.00 *gal*
Application event totals		65.51	9.57	110.75	2,299.95	

03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.25	11.43	132.29	1,819.71	1,935,000.00 *gal*
IW #3	Ground water	0.00	0.00	0.00	927.45	5,160,000.00 *gal*
Application event totals		78.25	11.43	132.29	2,747.16	

364 - 11/20/2002: Oats, silage-soft dough

Field name: 364

Crop: Oats, silage-soft dough

Plant date: 11/20/2002

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.98	8.83	89.89	1,407.64	2,700,000.00 *gal*
IW #3	Ground water	0.00	0.00	0.00	832.50	8,640,000.00 *gal*
Application event totals		55.98	8.83	89.89	2,240.14	

01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.66	7.84	90.71	1,247.75	2,475,000.00 *gal*
IW #3	Ground water	0.00	0.00	0.00	763.13	7,920,000.00 *gal*
Application event totals		53.66	7.84	90.71	2,010.88	

364 - 11/20/2002: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.39	9.40	108.85	1,497.30	2,970,000.00 *gal*
IW #3	Ground water	0.00	0.00	0.00	915.75	9,504,000.00 *gal*
Application event totals		64.39	9.40	108.85	2,413.05	

365 - 12/01/2019: Alfalfa, hay

Field name: 365

Crop: Alfalfa, hay

Plant date: 12/01/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	118.50	17.30	200.31	2,755.50	5,184,000.00 *gal*
IW #2	Ground water	0.00	0.00	0.00	1,655.18	13,824,000.00 *gal*
Application event totals		118.50	17.30	200.31	4,410.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,862.08	15,552,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,862.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,862.08	15,552,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,862.08	

365 - 12/01/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,655.18	13,824,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,655.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,655.18	13,824,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,655.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,793.12	14,976,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,793.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,448.29	12,096,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,448.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,448.29	12,096,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,448.29	

366 - 12/01/2019: Alfalfa, hay

Field name: 366

Crop: Alfalfa, hay

Plant date: 12/01/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

366 - 12/01/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.14	17.40	201.41	2,770.56	3,456,000.00 *gal*
IW #2	Ground water	0.00	0.00	0.00	2,080.29	11,520,000.00 *gal*
Application event totals		119.14	17.40	201.41	4,850.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,872.26	10,368,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,872.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,872.26	10,368,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,872.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,664.23	9,216,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,664.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,664.23	9,216,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,664.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,802.91	9,984,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,802.91	

366 - 12/01/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,456.20	8,064,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,456.20	

10/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #2	Ground water	0.00	0.00	0.00	1,456.20	8,064,000.00 *gal*
Application event totals		0.00	0.00	0.00	1,456.20	

367 - 11/20/2022: Oats, silage-soft dough

Field name: 367

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.19	11.86	120.73	1,890.57	2,430,000.00 *gal*
IW #1	Ground water	1.41	0.00	0.00	1,031.60	6,480,000.00 *gal*
Application event totals		76.59	11.86	120.73	2,922.16	

01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.34	10.42	120.60	1,658.90	2,205,000.00 *gal*
IW #1	Ground water	1.28	0.00	0.00	936.08	5,880,000.00 *gal*
Application event totals		72.61	10.42	120.60	2,594.97	

367 - 11/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.25	10.84	125.52	1,726.61	2,295,000.00 *gal*
IW #1	Ground water	1.33	0.00	0.00	974.29	6,120,000.00 *gal*
Application event totals		75.58	10.84	125.52	2,700.89	

367 - 06/19/2023: Corn, silage

Field name: 367

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	113.44	31.05	462.65	2,286.54	3,465,000.00 *gal*
Ponoche WD	Surface water	3.32	0.00	0.00	142.35	9,240,000.00 *gal*
Application event totals		116.76	31.05	462.65	2,428.90	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.62	0.00	0.00	155.29	10,080,000.00 *gal*
Application event totals		3.62	0.00	0.00	155.29	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.06	0.00	0.00	131.26	8,520,000.00 *gal*
Application event totals		3.06	0.00	0.00	131.26	

367 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.83	12.10	68.95	318.21	2,430,000.00 *gal*
Ponoche WD	Surface water	2.33	0.00	0.00	99.83	6,480,000.00 *gal*
Application event totals		31.16	12.10	68.95	418.05	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.58	0.00	0.00	153.45	9,960,000.00 *gal*
Application event totals		3.58	0.00	0.00	153.45	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.03	13.45	76.61	353.57	2,700,000.00 *gal*
Ponoche WD	Surface water	2.93	0.00	0.00	125.71	8,160,000.00 *gal*
Application event totals		34.96	13.45	76.61	479.29	

09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	2.59	0.00	0.00	110.92	7,200,000.00 *gal*
Application event totals		2.59	0.00	0.00	110.92	

10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	2.59	0.00	0.00	110.92	7,200,000.00 *gal*
Application event totals		2.59	0.00	0.00	110.92	

368 - 11/20/2022: Oats, silage-soft dough

Field name: 368

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

368 - 11/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.82	11.01	112.11	1,755.53	2,430,000.00 *gal*
IW #1	Ground water	1.31	0.00	0.00	957.91	6,480,000.00 *gal*
Application event totals		71.12	11.01	112.11	2,713.44	

01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.24	9.67	111.98	1,540.40	2,205,000.00 *gal*
IW #1	Ground water	1.18	0.00	0.00	869.22	5,880,000.00 *gal*
Application event totals		67.43	9.67	111.98	2,409.62	

03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.95	10.07	116.55	1,603.28	2,295,000.00 *gal*
IW #1	Ground water	1.23	0.00	0.00	904.69	6,120,000.00 *gal*
Application event totals		70.18	10.07	116.55	2,507.97	

368 - 06/19/2023: Corn, silage

Field name: 368

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.33	28.83	429.60	2,123.22	3,465,000.00 *gal*
Ponoche WD	Surface water	3.08	0.00	0.00	132.18	9,240,000.00 *gal*
Application event totals		108.42	28.83	429.60	2,255.40	

368 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.36	0.00	0.00	144.20	10,080,000.00 *gal*
Application event totals		3.36	0.00	0.00	144.20	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	2.84	0.00	0.00	121.88	8,520,000.00 *gal*
Application event totals		2.84	0.00	0.00	121.88	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.77	11.24	64.02	295.48	2,430,000.00 *gal*
Ponoche WD	Surface water	2.16	0.00	0.00	92.70	6,480,000.00 *gal*
Application event totals		28.93	11.24	64.02	388.19	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	3.32	0.00	0.00	142.48	9,960,000.00 *gal*
Application event totals		3.32	0.00	0.00	142.48	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.74	12.49	71.14	328.32	2,700,000.00 *gal*
Ponoche WD	Surface water	2.40	0.00	0.00	103.00	7,200,000.00 *gal*
Application event totals		32.14	12.49	71.14	431.32	

09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	2.72	0.00	0.00	116.73	8,160,000.00 *gal*
Application event totals		2.72	0.00	0.00	116.73	

368 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ponoche WD	Surface water	2.40	0.00	0.00	103.00	7,200,000.00 *gal*
Application event totals		2.40	0.00	0.00	103.00	

369 - 11/20/2002: Oats, silage-soft dough

Field name: 369

Crop: Oats, silage-soft dough

Plant date: 11/20/2002

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Fall Solid Manure	Corral solids	209.64	66.61	396.73	3,358.69	550.00 *ton*
Application event totals		209.64	66.61	396.73	3,358.69	

02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	2.19	0.00	0.00	1,604.70	10,080,000.00 *gal*
Application event totals		2.19	0.00	0.00	1,604.70	

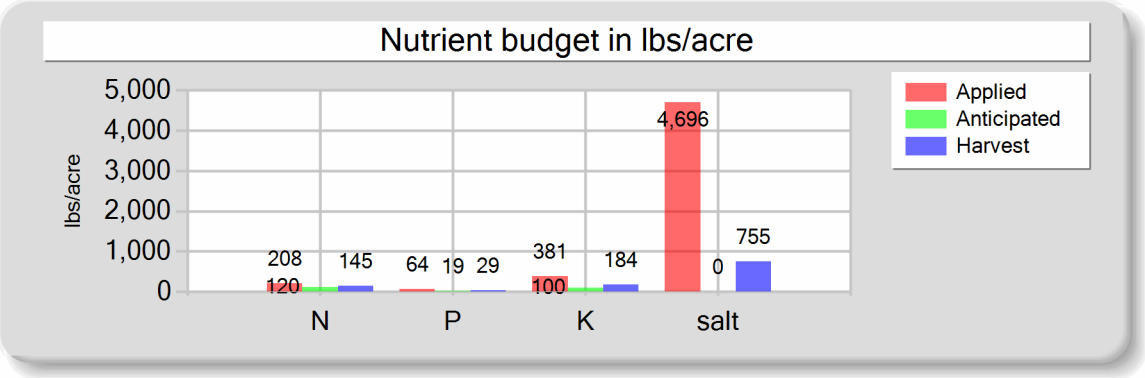
03/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	1.87	0.00	0.00	1,375.46	8,640,000.00 *gal*
Application event totals		1.87	0.00	0.00	1,375.46	

B. NUTRIENT BUDGET

350 - 11/20/2022: Oats, silage-soft dough

Field name: 350 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	201.30	63.96	380.95	3,225.11
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,470.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.30	63.96	380.95	4,695.66
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	144.55	29.40	183.75	755.00
Nutrient balance	63.75	34.56	197.20	3,940.66
Applied to removed ratio	1.44	2.18	2.07	6.22

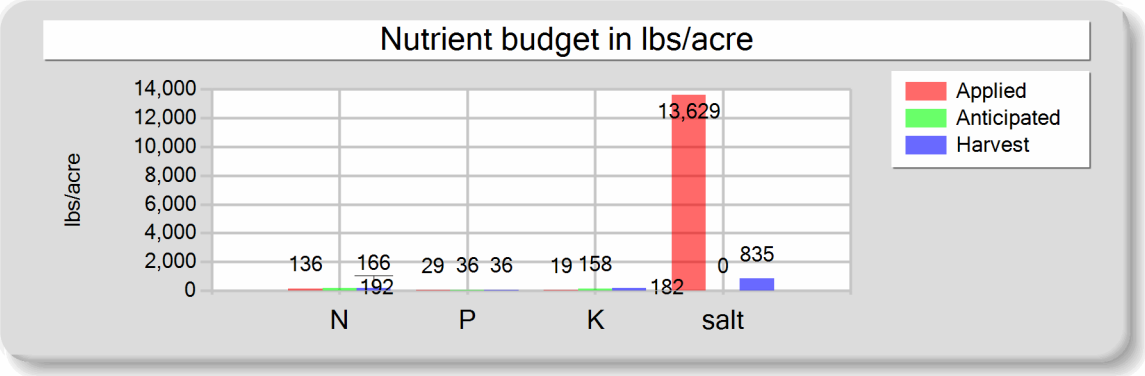
Fresh water applied	
10,680,000.00	*gallons*
393.31	*acre-inches*
4.92	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

350 - 07/03/2023: Corn, silage

Field name: 350 Crop: Corn, silage Plant date: 07/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	129.36	29.11	18.94	1,980.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	11,648.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	136.36	29.11	18.94	13,628.79
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	165.58	35.80	182.49	835.38
Nutrient balance	-29.22	-6.70	-163.55	12,793.41
Applied to removed ratio	0.82	0.81	0.10	16.31

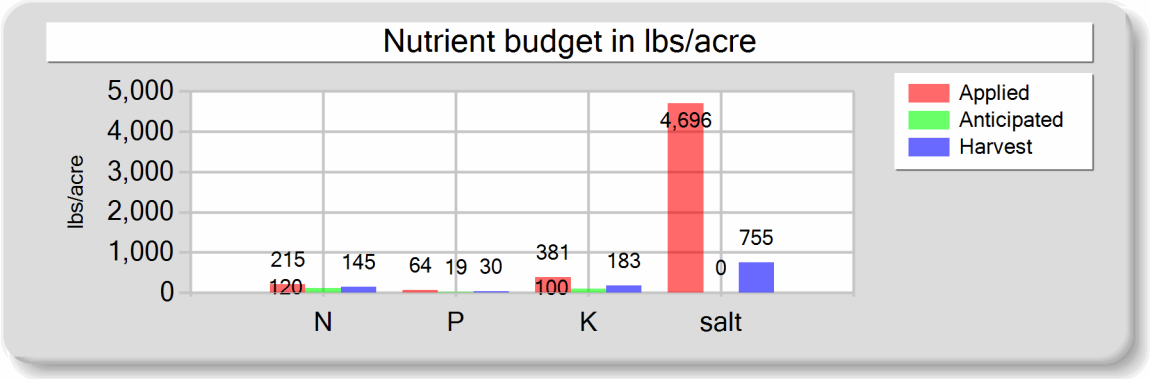
Fresh water applied	
84,600,000.00	*gallons*
3,115.53	*acre-inches*
38.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

351 - 11/20/2022: Oats, silage-soft dough

Field name: 351 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	201.30	63.96	380.95	3,225.11
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,470.56
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	215.30	63.96	380.95	4,695.66
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	145.29	29.89	183.26	755.00
Nutrient balance	70.01	34.07	197.69	3,940.66
Applied to removed ratio	1.48	2.14	2.08	6.22

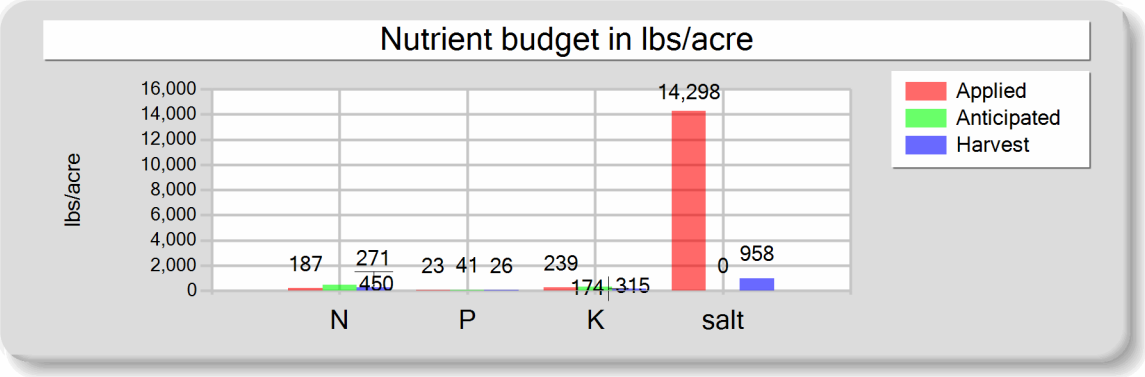
Fresh water applied	
10,680,000.00	*gallons*
393.31	*acre-inches*
4.92	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

352 - 11/01/2017: Alfalfa, hay

Field name: 352 Crop: Alfalfa, hay Plant date: 11/01/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	148.77	23.46	238.90	3,740.85
Fresh water	24.61	0.00	0.00	10,557.23
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	187.38	23.46	238.90	14,298.07
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	270.51	26.13	174.19	958.05
Nutrient balance	-83.13	-2.67	64.70	13,340.02
Applied to removed ratio	0.69	0.90	1.37	14.92

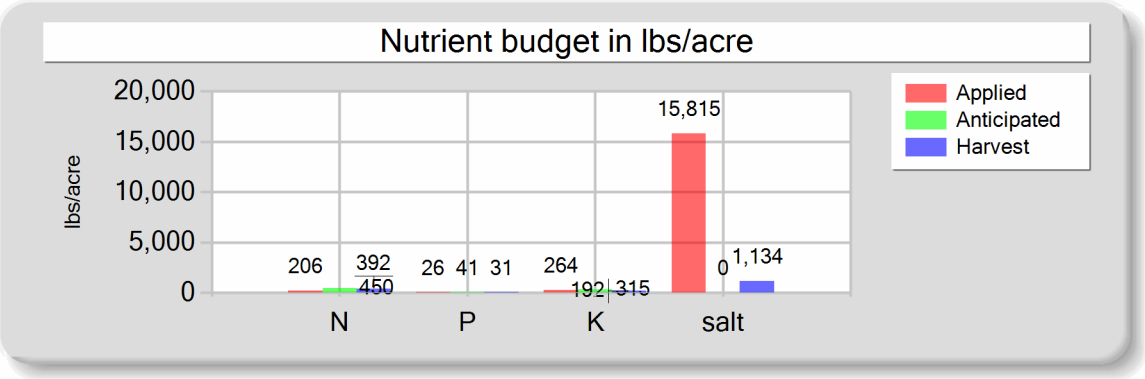
Fresh water applied	
71,040,000.00	*gallons*
2,616.16	*acre-inches*
35.84	*inches/acre*

Process wastewater applied	
5,400,000.00	*gallons*
198.86	*acre-inches*
2.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

353 - 11/01/2017: Alfalfa, hay

Field name: 353 Crop: Alfalfa, hay Plant date: 11/01/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	164.55	25.95	264.23	4,137.60
Fresh water	27.22	0.00	0.00	11,676.93
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	205.76	25.95	264.23	15,814.53
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	391.69	31.43	192.33	1,133.51
Nutrient balance	-185.93	-5.48	71.91	14,681.02
Applied to removed ratio	0.53	0.83	1.37	13.95

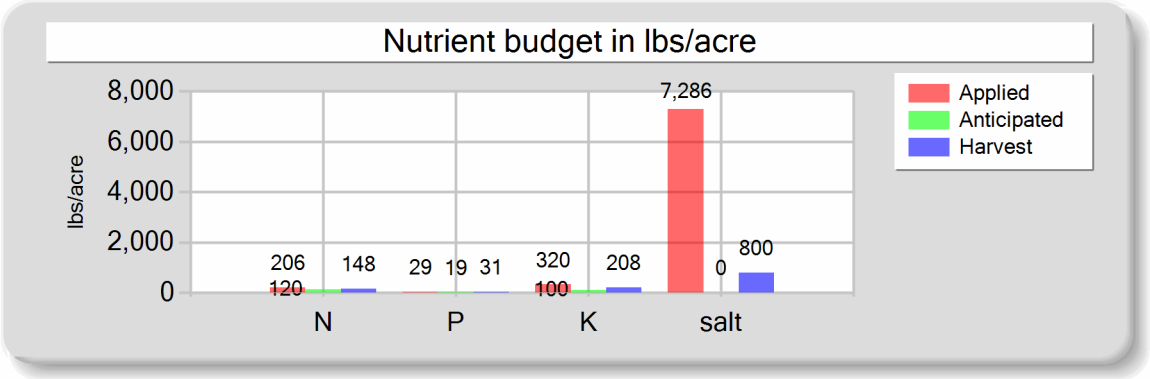
Fresh water applied	
71,040,000.00	*gallons*
2,616.16	*acre-inches*
39.64	*inches/acre*

Process wastewater applied	
5,400,000.00	*gallons*
198.86	*acre-inches*
3.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

354 - 11/20/2022: Oats, silage-soft dough

Field name: 354 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	192.63	28.86	320.35	4,597.16
Fresh water	6.27	0.00	0.00	2,689.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.89	28.86	320.35	7,286.17
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	148.07	31.20	207.66	800.26
Nutrient balance	57.82	-2.34	112.69	6,485.91
Applied to removed ratio	1.39	0.92	1.54	9.10

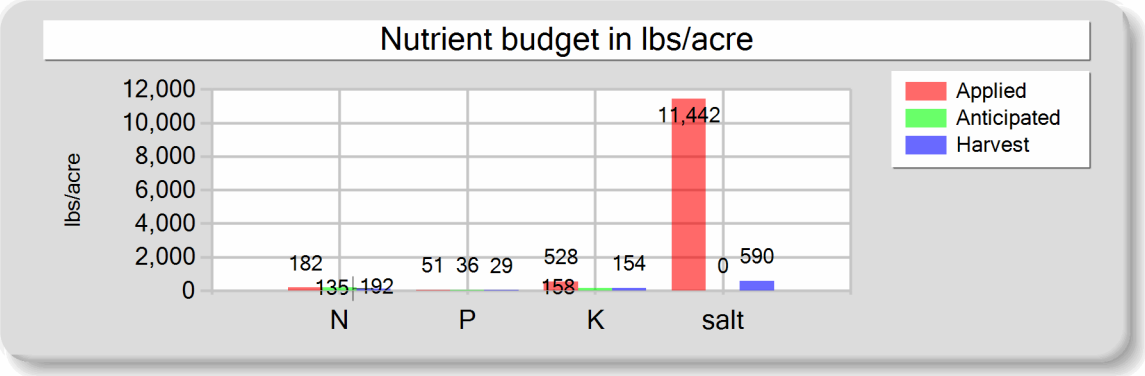
Fresh water applied	
15,120,000.00	*gallons*
556.82	*acre-inches*
9.13	*inches/acre*

Process wastewater applied	
5,670,000.00	*gallons*
208.81	*acre-inches*
3.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

354 - 06/23/2023: Corn, silage

Field name: 354 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	154.48	51.10	528.36	2,564.16
Fresh water	20.69	0.00	0.00	8,877.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	182.18	51.10	528.36	11,442.14
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	134.57	28.62	153.80	589.55
Nutrient balance	47.61	22.48	374.56	10,852.59
Applied to removed ratio	1.35	1.79	3.44	19.41

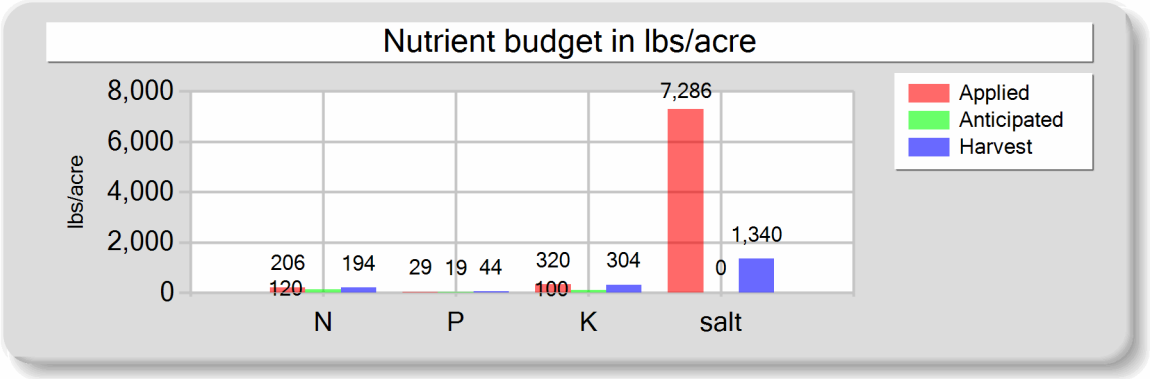
Fresh water applied	
49,920,000.00	*gallons*
1,838.38	*acre-inches*
30.14	*inches/acre*

Process wastewater applied	
7,470,000.00	*gallons*
275.09	*acre-inches*
4.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

355 - 11/20/2022: Oats, silage-soft dough

Field name: 355 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	192.63	28.86	320.35	4,597.16
Fresh water	6.27	0.00	0.00	2,689.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.89	28.86	320.35	7,286.17
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	194.10	43.99	304.33	1,340.17
Nutrient balance	11.79	-15.13	16.03	5,946.00
Applied to removed ratio	1.06	0.66	1.05	5.44

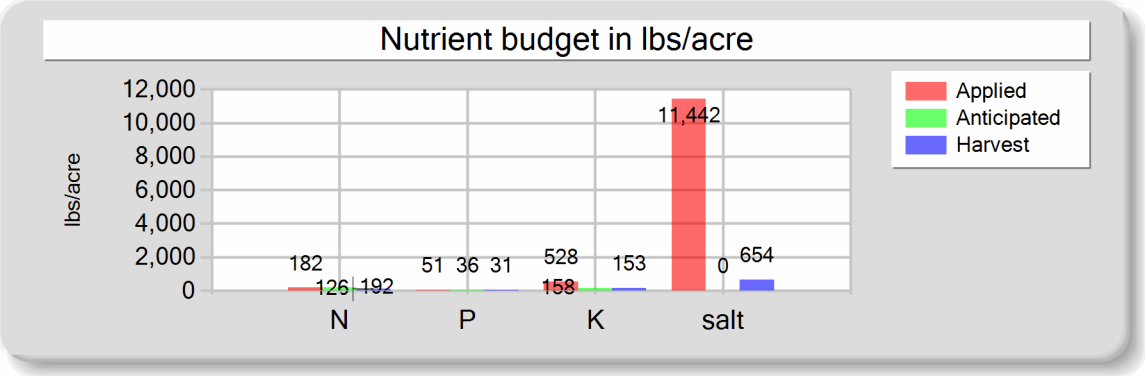
Fresh water applied	
15,120,000.00	*gallons*
556.82	*acre-inches*
9.13	*inches/acre*

Process wastewater applied	
5,670,000.00	*gallons*
208.81	*acre-inches*
3.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

355 - 06/23/2023: Corn, silage

Field name: 355 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	154.48	51.10	528.36	2,564.16
Fresh water	20.69	0.00	0.00	8,877.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	182.18	51.10	528.36	11,442.14
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	125.60	31.19	152.52	653.64
Nutrient balance	56.58	19.91	375.84	10,788.51
Applied to removed ratio	1.45	1.64	3.46	17.51

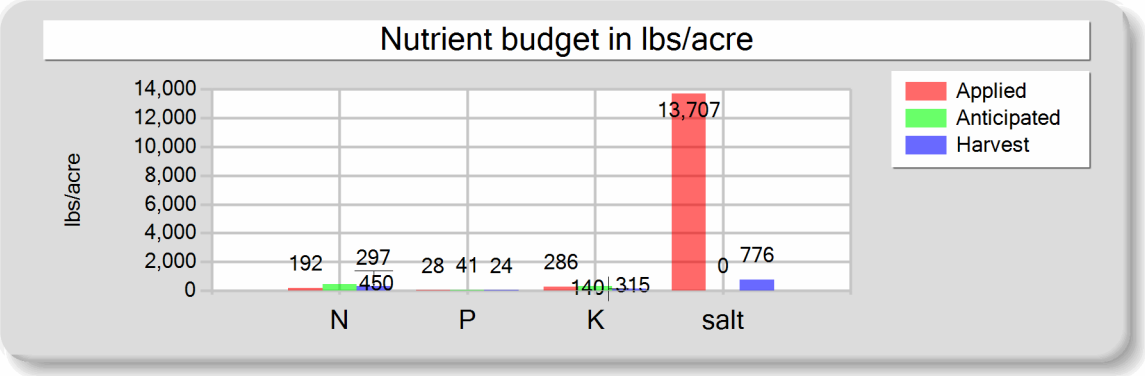
Fresh water applied	
49,920,000.00	*gallons*
1,838.38	*acre-inches*
30.14	*inches/acre*

Process wastewater applied	
7,470,000.00	*gallons*
275.09	*acre-inches*
4.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

356 - 12/01/2020: Alfalfa, hay

Field name: 356 Crop: Alfalfa, hay Plant date: 12/01/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	178.04	28.07	285.89	4,476.75
Fresh water	0.00	0.00	0.00	9,229.95
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	192.04	28.07	285.89	13,706.70
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	296.79	24.47	139.86	776.20
Nutrient balance	-104.75	3.60	146.04	12,930.50
Applied to removed ratio	0.65	1.15	2.04	17.66

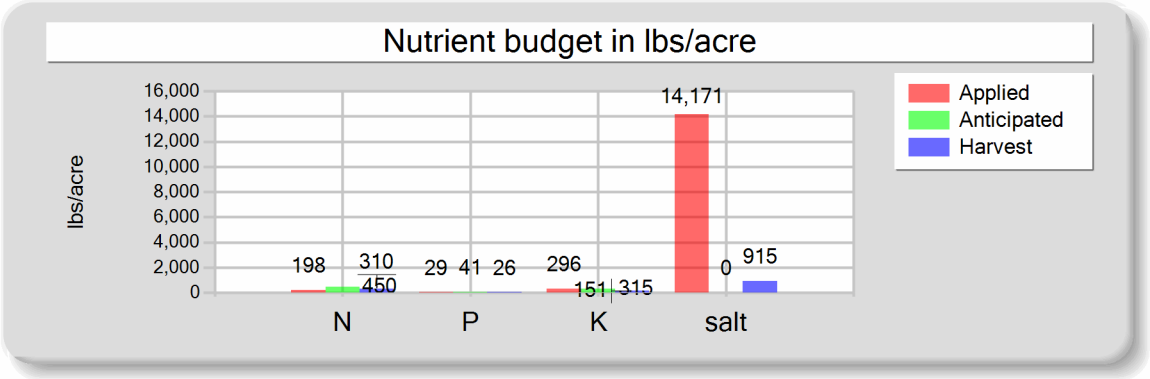
Fresh water applied	
60,240,000.00	*gallons*
2,218.43	*acre-inches*
36.37	*inches/acre*

Process wastewater applied	
5,400,000.00	*gallons*
198.86	*acre-inches*
3.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

357 - 12/01/2020: Alfalfa, hay

Field name: 357 Crop: Alfalfa, hay Plant date: 12/01/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	184.07	29.02	295.58	4,628.50
Fresh water	0.00	0.00	0.00	9,542.83
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	198.07	29.02	295.58	14,171.34
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	310.18	25.70	151.48	915.39
Nutrient balance	-112.11	3.33	144.10	13,255.95
Applied to removed ratio	0.64	1.13	1.95	15.48

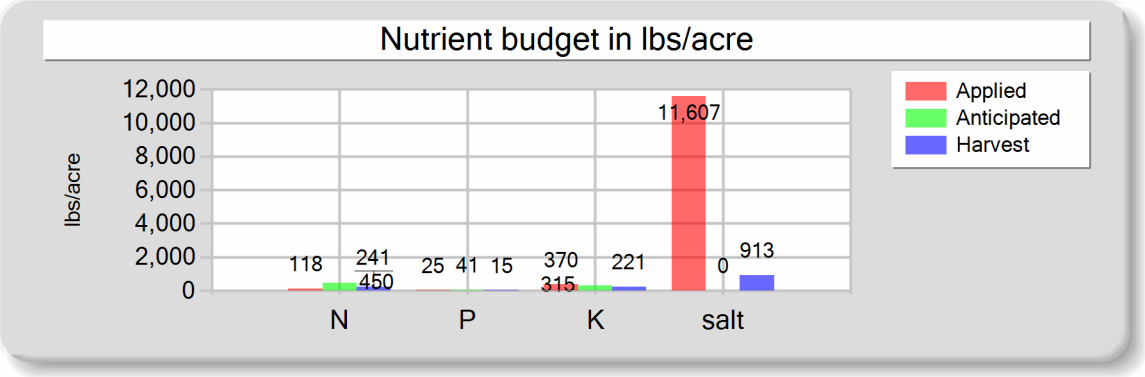
Fresh water applied	
60,240,000.00	*gallons*
2,218.43	*acre-inches*
37.60	*inches/acre*

Process wastewater applied	
5,400,000.00	*gallons*
198.86	*acre-inches*
3.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

358 - 12/01/2019: Alfalfa, hay

Field name: 358 Crop: Alfalfa, hay Plant date: 12/01/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	90.72	24.83	369.99	1,828.61
Fresh water	13.33	0.00	0.00	9,778.67
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	118.05	24.83	369.99	11,607.28
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	240.75	15.46	221.32	913.40
Nutrient balance	-122.70	9.37	148.67	10,693.88
Applied to removed ratio	0.49	1.61	1.67	12.71

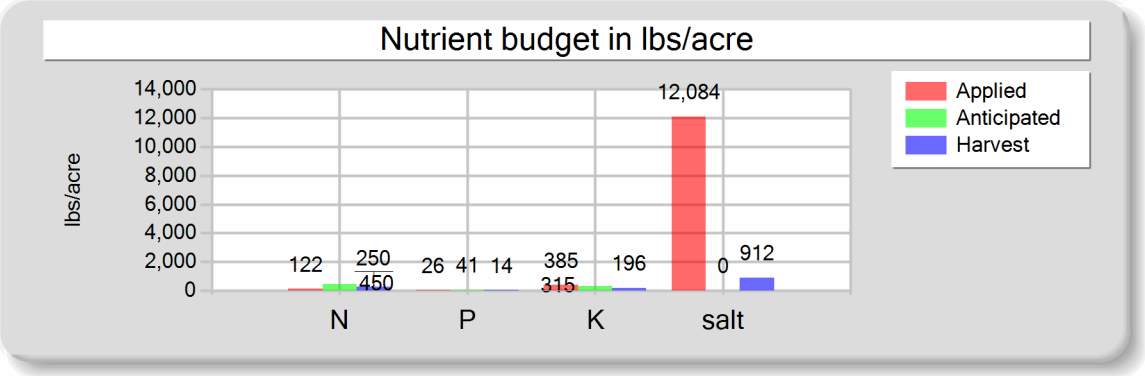
Fresh water applied	
71,820,000.00	*gallons*
2,644.89	*acre-inches*
34.80	*inches/acre*

Process wastewater applied	
3,240,000.00	*gallons*
119.32	*acre-inches*
1.57	*inches/acre*

Total harvests for the crop	
1	*harvests*

359 - 12/01/2019: Alfalfa, hay

Field name: 359 Crop: Alfalfa, hay Plant date: 12/01/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	94.45	25.85	385.20	1,903.76
Fresh water	13.88	0.00	0.00	10,180.53
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	122.32	25.85	385.20	12,084.29
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	249.96	13.55	195.77	911.57
Nutrient balance	-127.63	12.31	189.43	11,172.72
Applied to removed ratio	0.49	1.91	1.97	13.26

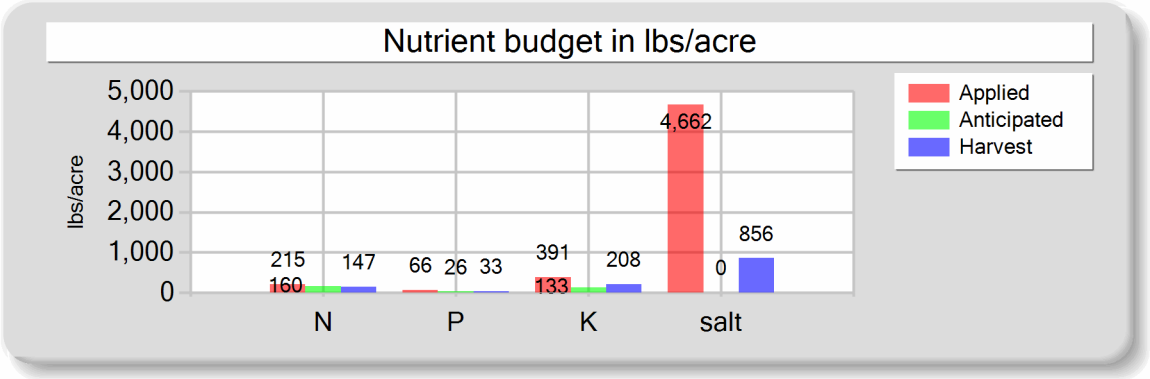
Fresh water applied	
71,820,000.00	*gallons*
2,644.89	*acre-inches*
36.23	*inches/acre*

Process wastewater applied	
3,240,000.00	*gallons*
119.32	*acre-inches*
1.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

360 - 11/20/2022: Oats, silage-soft dough

Field name: 360 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	206.46	65.60	390.72	3,307.80
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.85	0.00	0.00	1,354.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.31	65.60	390.72	4,662.42
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	146.70	33.33	207.84	856.44
Nutrient balance	68.61	32.27	182.88	3,805.98
Applied to removed ratio	1.47	1.97	1.88	5.44

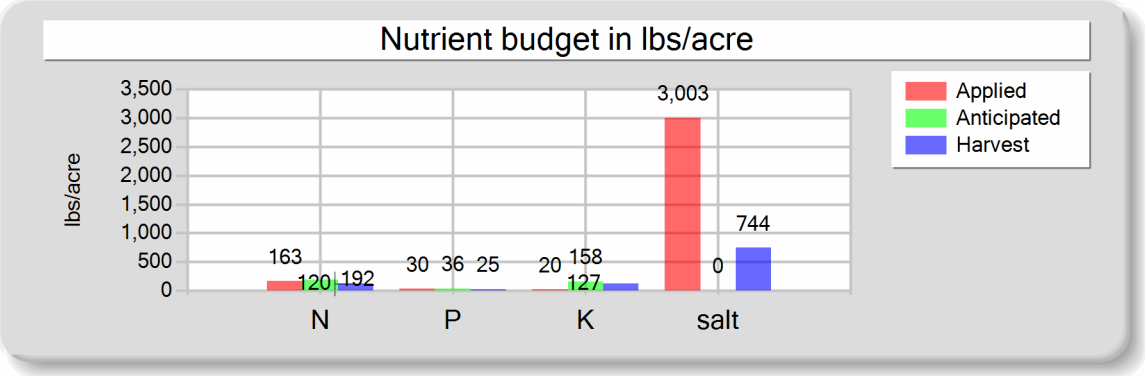
Fresh water applied	
8,640,000.00	*gallons*
318.18	*acre-inches*
4.82	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

360 - 06/22/2023: Corn, silage

Field name: 360 Crop: Corn, silage Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	133.28	29.99	19.52	2,040.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	22.47	0.00	0.00	963.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	162.75	29.99	19.52	3,003.16
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	119.68	25.16	126.51	743.95
Nutrient balance	43.08	4.83	-106.99	2,259.22
Applied to removed ratio	1.36	1.19	0.15	4.04

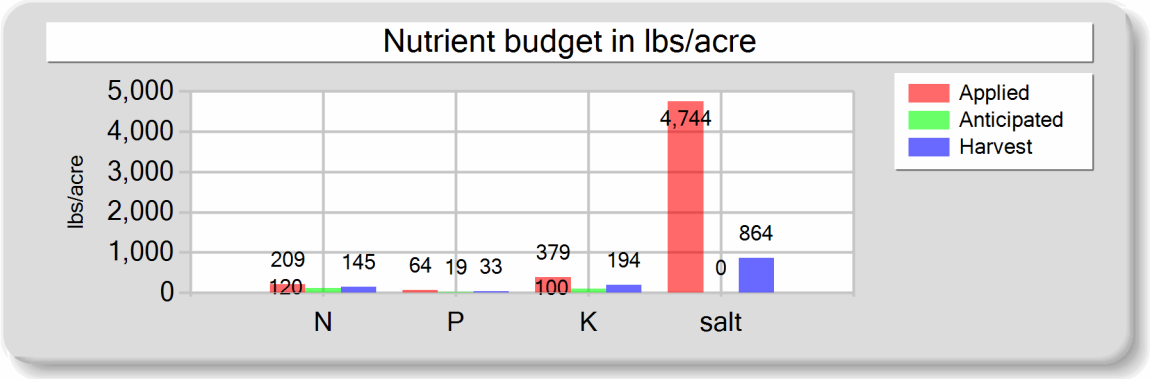
Fresh water applied	
63,480,000.00	*gallons*
2,337.75	*acre-inches*
35.42	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

361 - 11/20/2022: Oats, silage-soft dough

Field name: 361 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	200.39	63.67	379.23	3,210.51
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.09	0.00	0.00	1,533.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.48	63.67	379.23	4,744.42
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	145.18	33.11	193.58	864.14
Nutrient balance	64.30	30.56	185.65	3,880.29
Applied to removed ratio	1.44	1.92	1.96	5.49

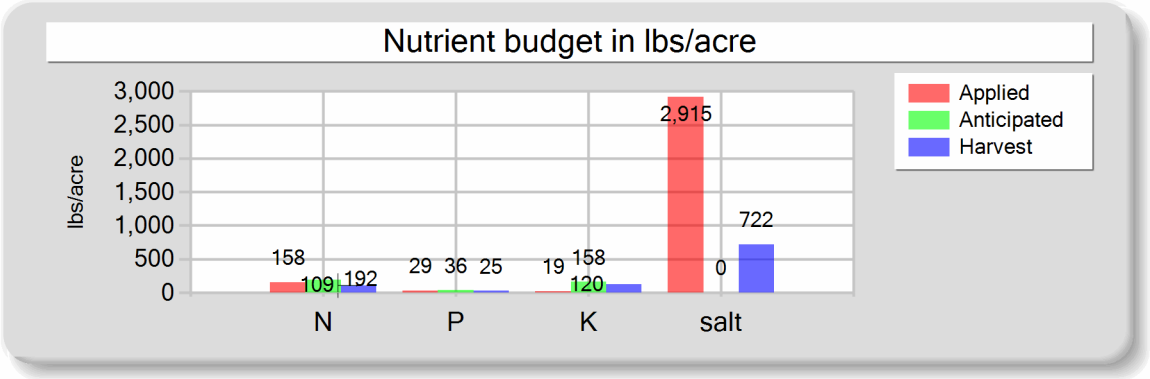
Fresh water applied	
10,080,000.00	*gallons*
371.21	*acre-inches*
5.46	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

361 - 06/22/2023: Corn, silage

Field name: 361 Crop: Corn, silage Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	129.36	29.11	18.94	1,980.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	21.81	0.00	0.00	934.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	158.17	29.11	18.94	2,914.84
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	108.83	25.46	120.34	722.06
Nutrient balance	49.34	3.64	-101.40	2,192.77
Applied to removed ratio	1.45	1.14	0.16	4.04

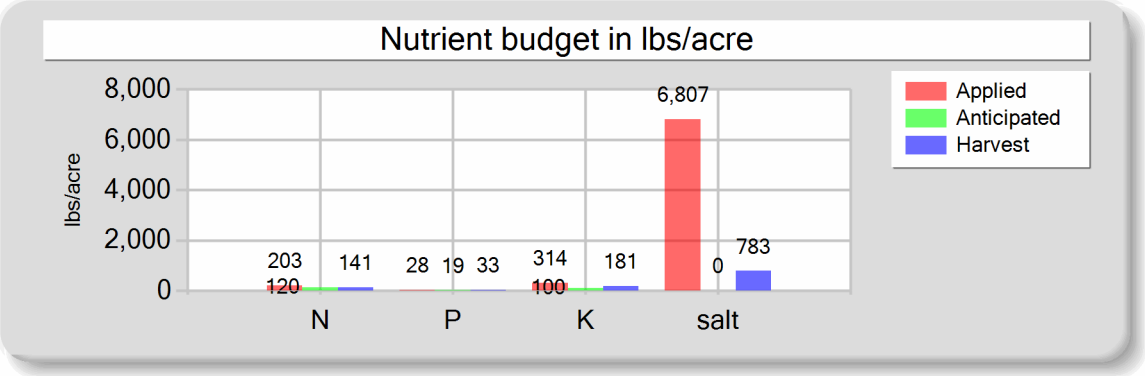
Fresh water applied	
63,480,000.00	*gallons*
2,337.75	*acre-inches*
34.38	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

362 - 11/20/2022: Oats, silage-soft dough

Field name: 362 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	189.25	28.42	314.24	4,527.86
Fresh water	0.00	0.00	0.00	2,279.32
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	203.25	28.42	314.24	6,807.18
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	140.75	32.96	180.51	782.89
Nutrient balance	62.50	-4.54	133.73	6,024.29
Applied to removed ratio	1.44	0.86	1.74	8.69

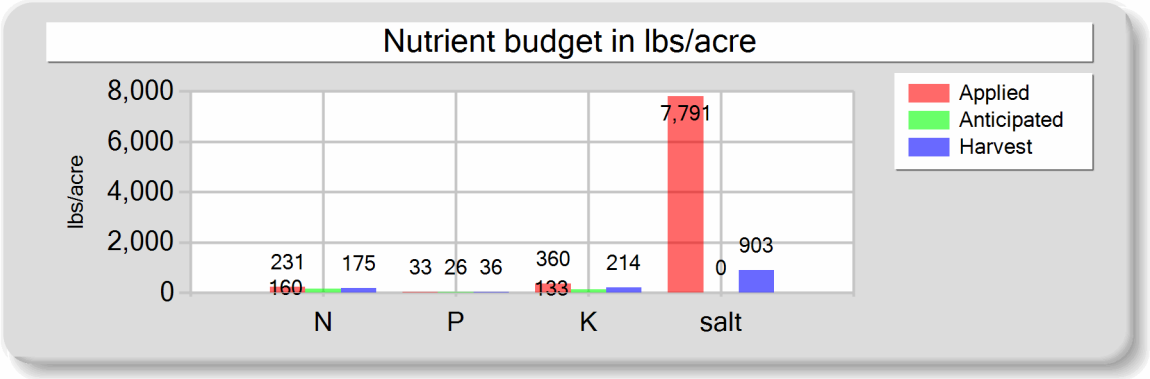
Fresh water applied	
15,120,000.00	*gallons*
556.82	*acre-inches*
8.98	*inches/acre*

Process wastewater applied	
5,670,000.00	*gallons*
208.81	*acre-inches*
3.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

363 - 11/20/2022: Oats, silage-soft dough

Field name: 363 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	216.87	32.52	360.42	5,181.23
Fresh water	0.00	0.00	0.00	2,609.80
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	230.87	32.52	360.42	7,791.04
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	175.48	36.06	213.94	902.73
Nutrient balance	55.38	-3.54	146.48	6,888.30
Applied to removed ratio	1.32	0.90	1.68	8.63

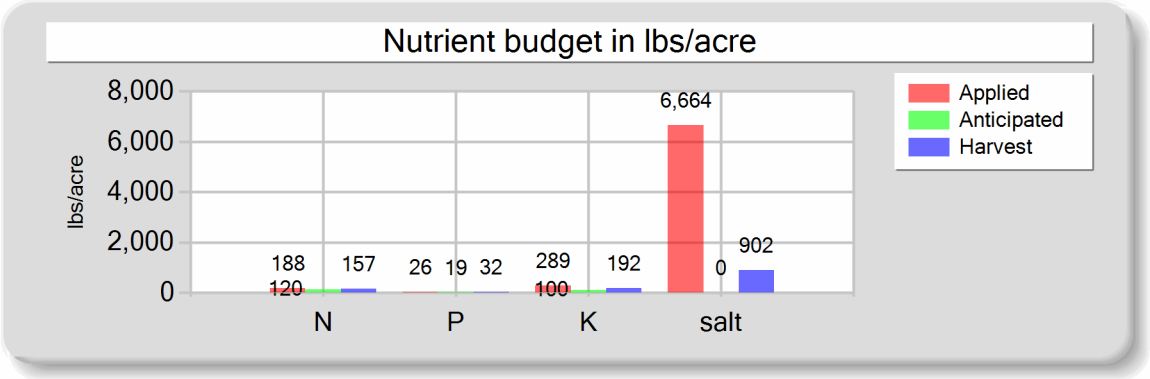
Fresh water applied	
14,520,000.00	*gallons*
534.72	*acre-inches*
10.28	*inches/acre*

Process wastewater applied	
5,445,000.00	*gallons*
200.52	*acre-inches*
3.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

364 - 11/20/2002: Oats, silage-soft dough

Field name: 364 Crop: Oats, silage-soft dough Plant date: 11/20/2002



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	174.03	26.07	289.45	4,152.69
Fresh water	0.00	0.00	0.00	2,511.39
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	188.03	26.07	289.45	6,664.07
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	157.10	32.28	191.65	902.05
Nutrient balance	30.93	-6.21	97.80	5,762.02
Applied to removed ratio	1.20	0.81	1.51	7.39

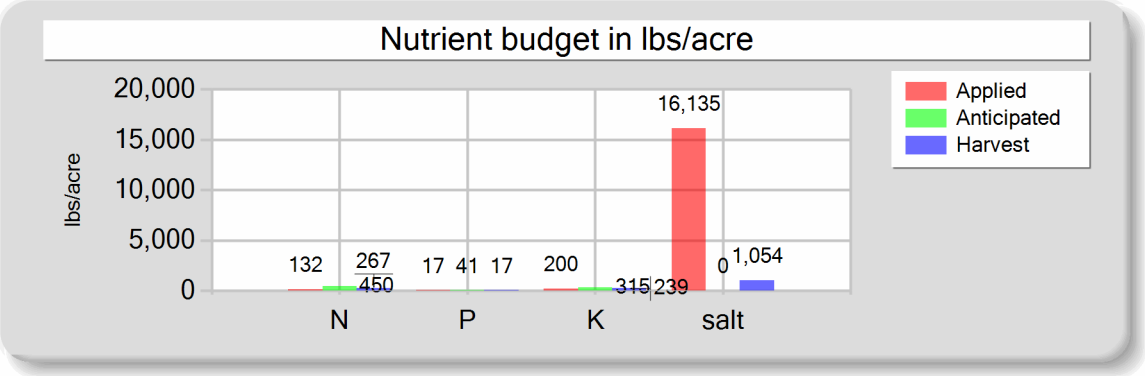
Fresh water applied	
26,064,000.00	*gallons*
959.85	*acre-inches*
9.90	*inches/acre*

Process wastewater applied	
8,145,000.00	*gallons*
299.95	*acre-inches*
3.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

365 - 12/01/2019: Alfalfa, hay

Field name: 365 Crop: Alfalfa, hay Plant date: 12/01/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	118.50	17.30	200.31	2,755.50
Fresh water	0.00	0.00	0.00	13,379.40
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	132.50	17.30	200.31	16,134.91
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	267.16	16.89	239.40	1,054.22
Nutrient balance	-134.66	0.42	-39.09	15,080.69
Applied to removed ratio	0.50	1.02	0.84	15.31

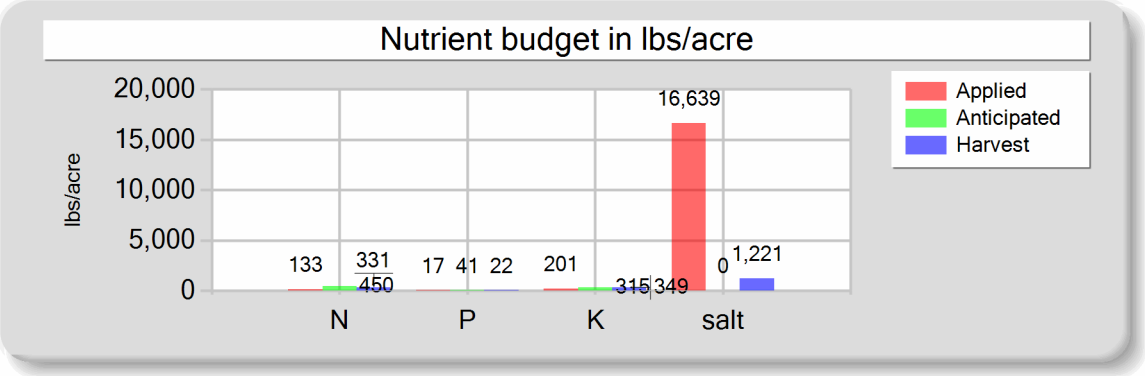
Fresh water applied	
111,744,000.00	*gallons*
4,115.15	*acre-inches*
44.73	*inches/acre*

Process wastewater applied	
5,184,000.00	*gallons*
190.91	*acre-inches*
2.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

366 - 12/01/2019: Alfalfa, hay

Field name: 366 Crop: Alfalfa, hay Plant date: 12/01/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	119.14	17.40	201.41	2,770.56
Fresh water	0.00	0.00	0.00	13,868.57
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	133.14	17.40	201.41	16,639.13
Anticipated crop nutrient removal	450.00	40.50	315.00	0.00
Actual crop nutrient removal	330.94	21.58	348.83	1,221.36
Nutrient balance	-197.79	-4.18	-147.42	15,417.77
Applied to removed ratio	0.40	0.81	0.58	13.62

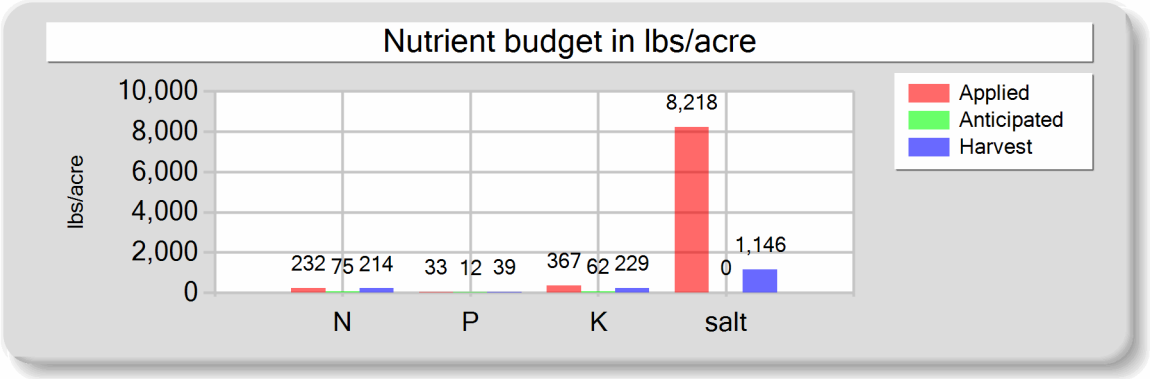
Fresh water applied	
76,800,000.00	*gallons*
2,828.28	*acre-inches*
46.37	*inches/acre*

Process wastewater applied	
3,456,000.00	*gallons*
127.27	*acre-inches*
2.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

367 - 11/20/2022: Oats, silage-soft dough

Field name: 367 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	220.77	33.12	366.85	5,276.07
Fresh water	4.01	0.00	0.00	2,941.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	231.78	33.12	366.85	8,218.03
Anticipated crop nutrient removal	75.00	12.00	62.25	0.00
Actual crop nutrient removal	214.15	39.14	229.29	1,145.81
Nutrient balance	17.63	-6.02	137.56	7,072.22
Applied to removed ratio	1.08	0.85	1.60	7.17

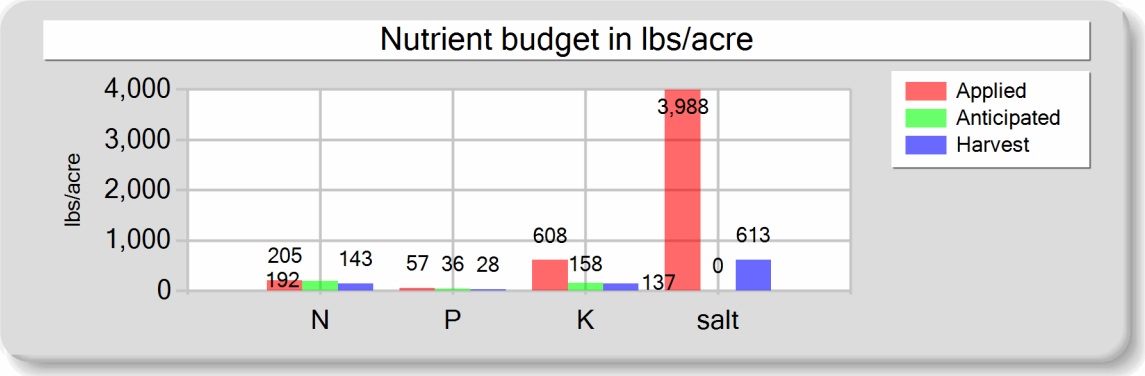
Fresh water applied	
18,480,000.00	*gallons*
680.56	*acre-inches*
10.47	*inches/acre*

Process wastewater applied	
6,930,000.00	*gallons*
255.21	*acre-inches*
3.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

367 - 06/19/2023: Corn, silage

Field name: 367 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	174.29	56.60	608.20	2,958.33
Fresh water	24.03	0.00	0.00	1,029.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.32	56.60	608.20	3,988.08
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	143.37	28.24	136.86	612.59
Nutrient balance	61.95	28.36	471.34	3,375.48
Applied to removed ratio	1.43	2.00	4.44	6.51

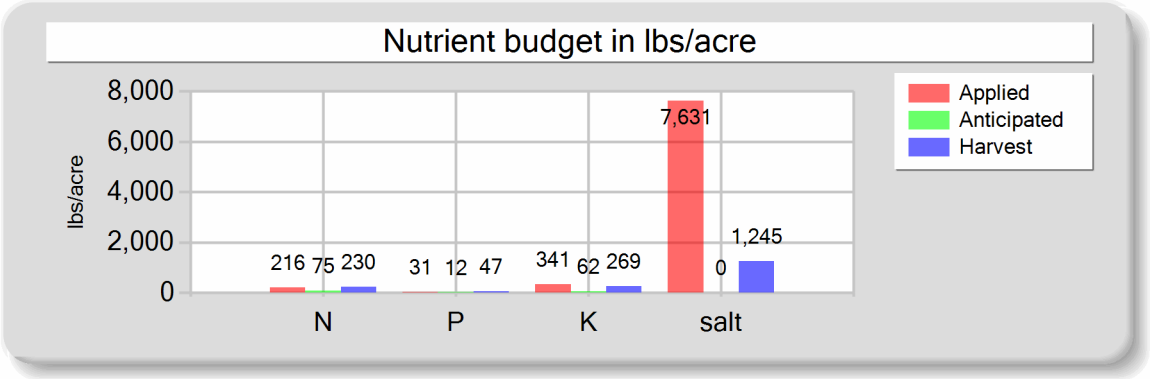
Fresh water applied	
66,840,000.00	*gallons*
2,461.49	*acre-inches*
37.87	*inches/acre*

Process wastewater applied	
8,595,000.00	*gallons*
316.52	*acre-inches*
4.87	*inches/acre*

Total harvests for the crop	
1	*harvests*

368 - 11/20/2022: Oats, silage-soft dough

Field name: 368 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	205.00	30.75	340.64	4,899.21
Fresh water	3.72	0.00	0.00	2,731.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.73	30.75	340.64	7,631.03
Anticipated crop nutrient removal	75.00	12.00	62.25	0.00
Actual crop nutrient removal	230.04	46.80	268.92	1,245.08
Nutrient balance	-14.31	-16.05	71.72	6,385.94
Applied to removed ratio	0.94	0.66	1.27	6.13

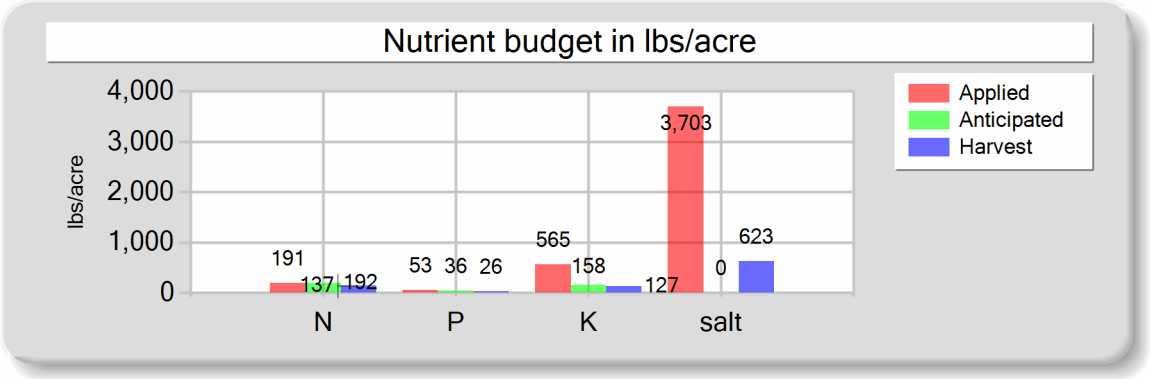
Fresh water applied	
18,480,000.00	*gallons*
680.56	*acre-inches*
9.72	*inches/acre*

Process wastewater applied	
6,930,000.00	*gallons*
255.21	*acre-inches*
3.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

368 - 06/19/2023: Corn, silage

Field name: 368 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	161.84	52.56	564.76	2,747.02
Fresh water	22.31	0.00	0.00	956.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.15	52.56	564.76	3,703.21
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	137.13	25.63	126.74	623.31
Nutrient balance	54.03	26.94	438.02	3,079.90
Applied to removed ratio	1.39	2.05	4.46	5.94

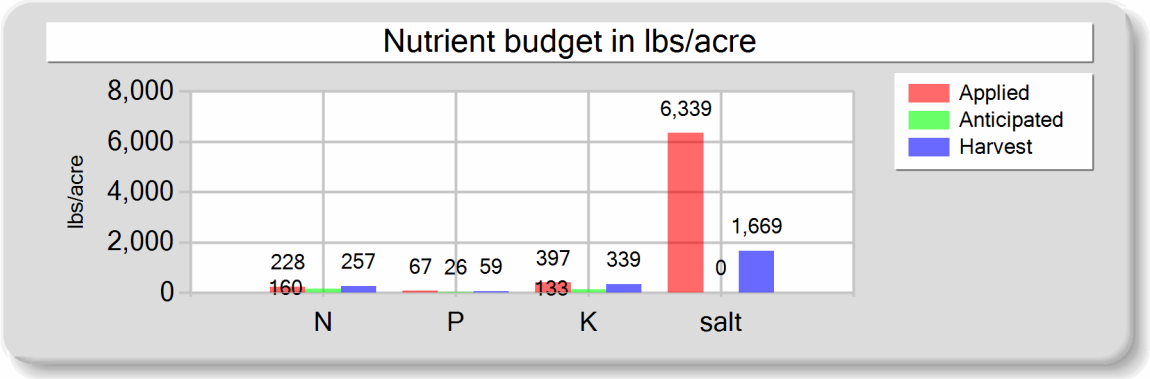
Fresh water applied	
66,840,000.00	*gallons*
2,461.49	*acre-inches*
35.16	*inches/acre*

Process wastewater applied	
8,595,000.00	*gallons*
316.52	*acre-inches*
4.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

369 - 11/20/2002: Oats, silage-soft dough

Field name: 369 Crop: Oats, silage-soft dough Plant date: 11/20/2002



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	209.64	66.61	396.73	3,358.69
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	4.06	0.00	0.00	2,980.17
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	227.70	66.61	396.73	6,338.86
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	257.35	59.42	339.37	1,668.89
Nutrient balance	-29.66	7.19	57.36	4,669.97
Applied to removed ratio	0.88	1.12	1.17	3.80

Fresh water applied	
18,720,000.00	*gallons*
689.39	*acre-inches*
10.61	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall Solid Manure

Sample and source description: Fall Solid Manure

Sample date: 10/07/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 33.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,600.00	5,910.00	35,200.00							29.80
DL	100.00	100.00	100.00							0.01

Spring Solid Manure

Sample and source description: Spring Solid Manure

Sample date: 05/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,600.00	4,410.00	2,870.00	15,100.00	5,440.00	2,190.00	3,200.00	989.00		30.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 12/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	241.00	163.00	0.00	0.00	38.00	387.00								4,600.00	6,060
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 03/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	252.00	165.00	0.00	0.00	36.80	426.00								4,680.00	5,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	255.00	148.00	0.00	0.00	69.80	1,040.00								3,660.00	5,140
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 09/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	92.40	58.80	0.00	0.00	38.80	221.00								2,220.00	1,020
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

IW #1

IW #1

IW 1

Sample description: IW 1

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.69	0.00	1.69								2,220.00	1,240
DL	0.50	0.50	0.50								1.00	10

IW #2

IW 2

Sample description: IW 2

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								2,270.00	1,320
DL	0.50	0.50	0.50								1.00	10

IW #3

IW 3

Sample description: IW 3

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								2,020.00	1,120
DL	0.50	0.50	0.50								1.00	10

IW #4

IW #4

IW 4

Sample description: IW 4

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.03	0.00	0.00								2,270.00	1,300
DL	0.50	0.50	0.50								1.00	10

Ponoche WD

Ponoche WD

Sample description: Ponoche WD

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.80	0.00	0.00								247.00	120
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

350

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			31.00				
DL			4.00				

351

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			13.00				
DL			4.00				

352

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			57.00				
DL			4.00				

353

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			46.00				
DL			4.00				

354

354

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			29.00				
DL			4.00				

355

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			31.00				
DL			4.00				

356

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			19.00				
DL			4.00				

357

357

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			17.00				
DL			4.00				

358

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			38.00				
DL			4.00				

359

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			7.00				
DL			4.00				

360

360

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			9.00				
DL			4.00				

361

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			8.00				
DL			4.00				

362

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00				
DL			4.00				

363

363

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			30.00				
DL			4.00				

364

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			36.00				
DL			4.00				

365

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			41.00				
DL			4.00				

366

366

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			42.00				
DL			4.00				

367

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			48.00				
DL			4.00				

368

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			31.00				
DL			4.00				

369

369

Soil

Sample and source description: Soil

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			14.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

350 - 07/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,330.00	720.00	3,670.00		5.60
DL	100.00	100.00	100.00		0.01

351 - 11/20/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,930.00	1,220.00	7,480.00		7.39
DL	100.00	100.00	100.00		0.01

352 - 11/01/2017: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 15.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,400.00	2,550.00	17,000.00		11.00
DL	100.00	100.00	100.00		0.01

353 - 11/01/2017: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 13.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,400.00	2,680.00	16,400.00		11.20
DL	100.00	100.00	100.00		0.01

354 - 11/20/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,790.00	1,220.00	8,120.00		8.79
DL	100.00	100.00	100.00		0.01

354 - 06/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,150.00	670.00	3,600.00		4.60
DL	100.00	100.00	100.00		0.01

355 - 11/20/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,590.00	1,720.00	11,900.00		13.20
DL	100.00	100.00	100.00		0.01

355 - 06/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,940.00	730.00	3,570.00		5.10
DL	100.00	100.00	100.00		0.01

356 - 12/01/2020: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 14.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	36,500.00	3,010.00	17,200.00		11.10
DL	100.00	100.00	100.00		0.01

357 - 12/01/2020: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 16.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,400.00	2,850.00	16,800.00		12.10
DL	100.00	100.00	100.00		0.01

358 - 12/01/2019: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 12.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,500.00	1,830.00	26,200.00		12.40
DL	100.00	100.00	100.00		0.01

359 - 12/01/2019: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 13.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,600.00	1,550.00	22,400.00		12.10
DL	100.00	100.00	100.00		0.01

360 - 11/20/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,590.00	1,270.00	7,920.00		9.83
DL	100.00	100.00	100.00		0.01

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,700.00	1,280.00	7,580.00		9.53
DL	100.00	100.00	100.00		0.01

360 - 06/22/2023: Corn, silage

360 - 06/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,330.00	700.00	3,520.00		6.90
DL	100.00	100.00	100.00		0.01

361 - 06/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,120.00	730.00	3,450.00		6.90
DL	100.00	100.00	100.00		0.01

362 - 11/20/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,380.00	1,260.00	6,900.00		8.88
DL	100.00	100.00	100.00		0.01

364 - 11/20/2002: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,230.00	1,280.00	7,600.00		9.34
DL	100.00	100.00	100.00		0.01

365 - 12/01/2019: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 23.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,100.00	1,460.00	20,700.00		11.90
DL	100.00	100.00	100.00		0.01

366 - 12/01/2019: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,600.00	1,930.00	31,200.00		11.90
DL	100.00	100.00	100.00		0.01

367 - 11/20/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,800.00	1,060.00	6,210.00		9.49
DL	100.00	100.00	100.00		0.01

367 - 06/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,300.00	650.00	3,150.00		4.70
DL	100.00	100.00	100.00		0.01

368 - 11/20/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,390.00	1,300.00	7,470.00		8.96
DL	100.00	100.00	100.00		0.01

368 - 06/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,960.00	740.00	3,660.00		6.00
DL	100.00	100.00	100.00		0.01

369 - 11/20/2002: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,150.00	1,420.00	8,110.00		9.61
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

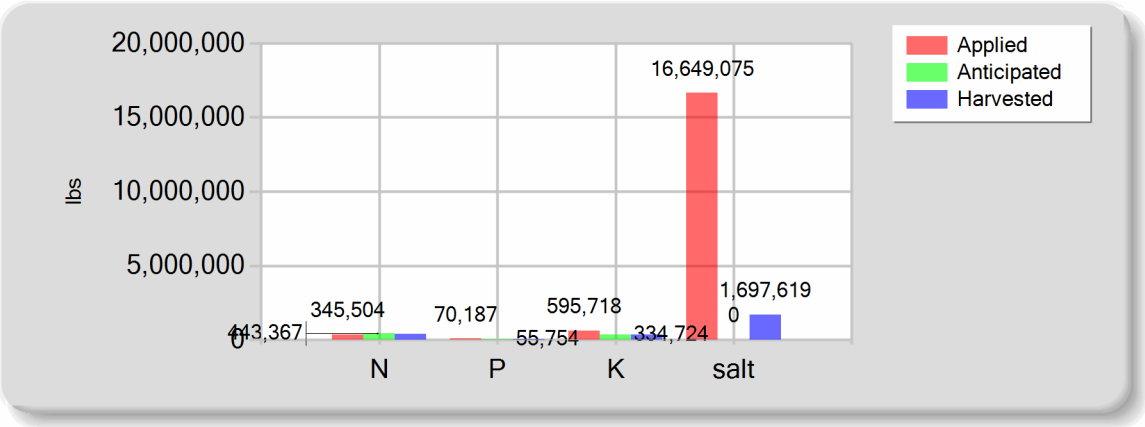
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

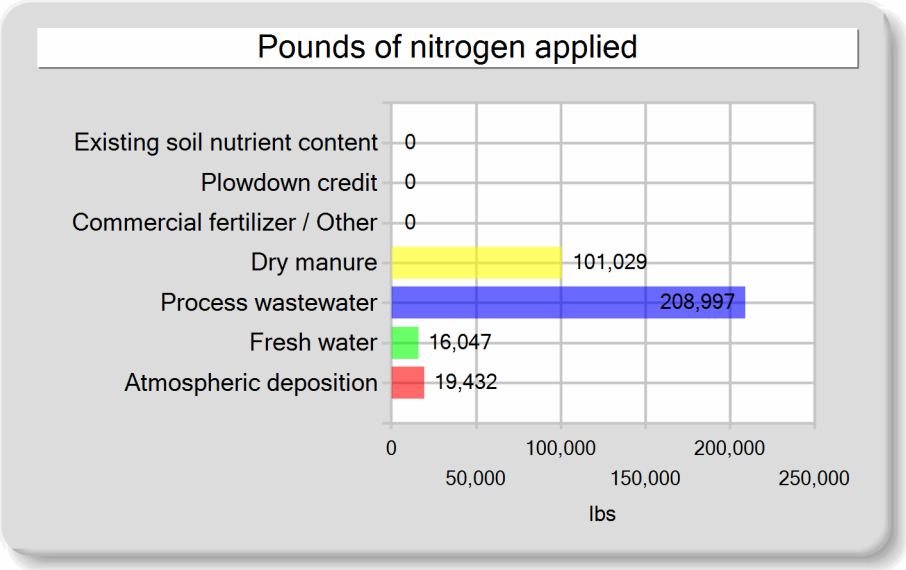
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

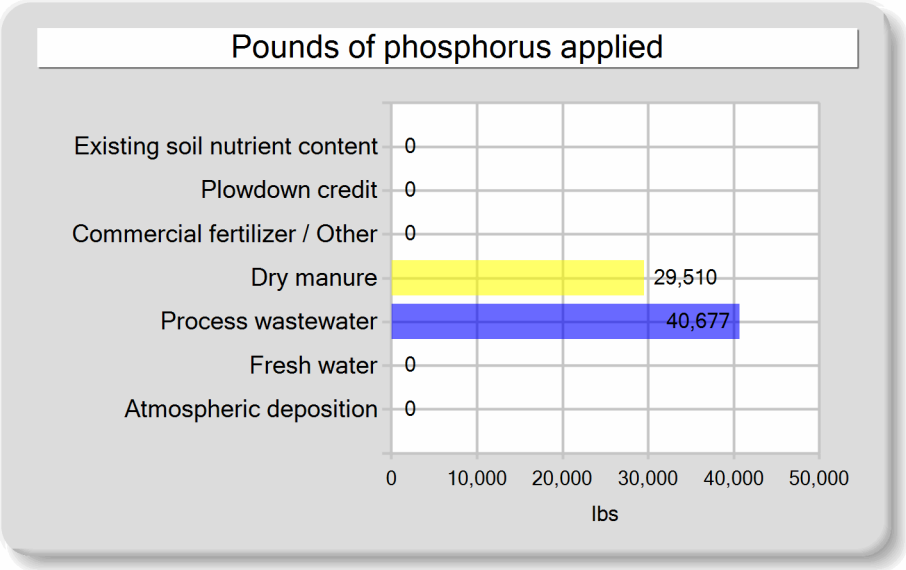
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	101,028.60	29,509.65	142,406.35	1,598,641.20
Process wastewater	208,996.94	40,677.02	453,311.25	4,689,901.68
Fresh water	16,046.80	0.00	0.00	10,360,532.47
Atmospheric deposition	19,432.00	0.00	0.00	0.00
Total nutrients applied	345,504.34	70,186.67	595,717.60	16,649,075.35
Anticipated crop nutrient removal	443,367.00	55,754.10	334,723.95	0.00
Actual crop nutrient removal	370,102.54	56,071.06	369,044.40	1,697,619.48
Nutrient balance	-24,598.20	14,115.61	226,673.20	14,951,455.87
Applied to removed ratio	0.93	1.25	1.61	9.81

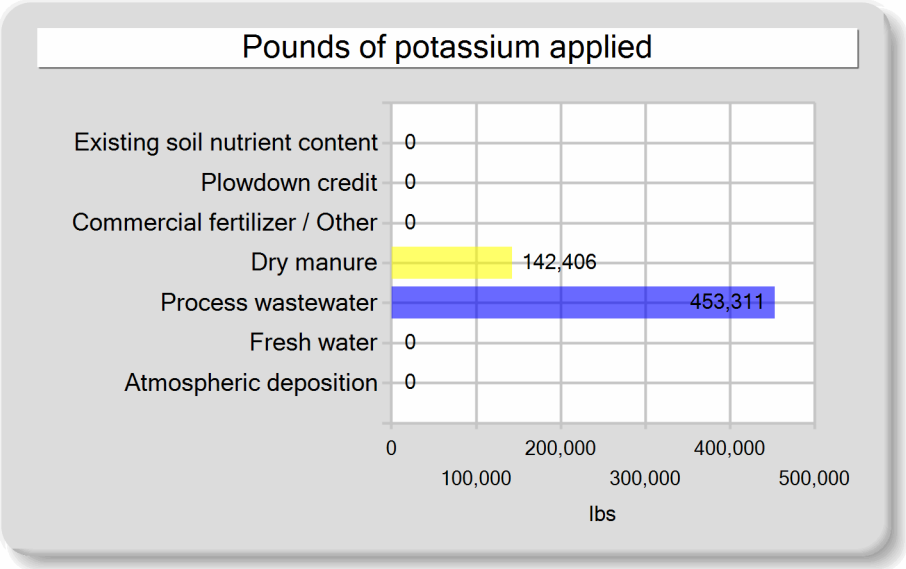
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

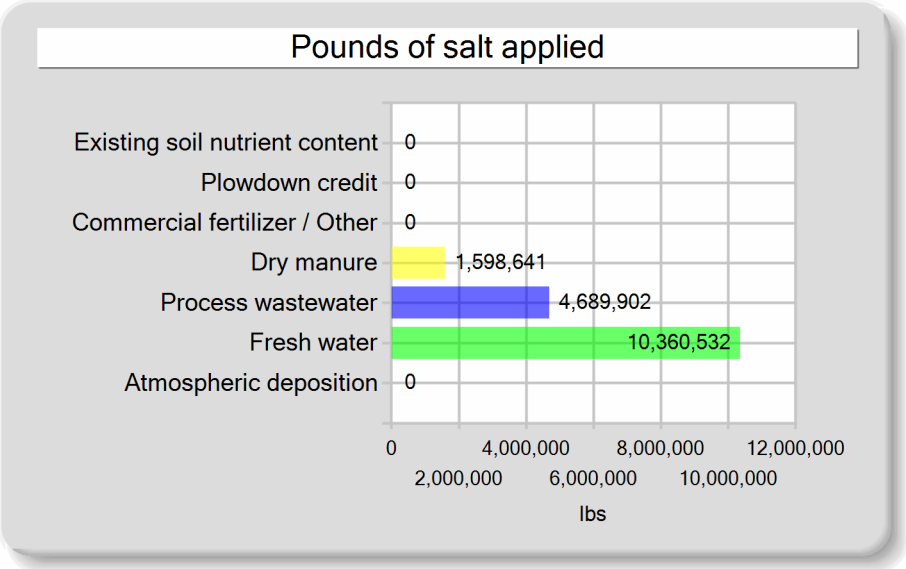


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Brad Nyman
Brad Nyman (Jun 22, 2024 10:30 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Brad Nyman	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 22, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Live Oak Dairy F
PO Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/28/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 1	T3240027A	Electrical Conductivity (EC)	EPA 120.1 and	0.226	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	145	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.8	95.2 (T3240023E)	96.4 (T3240023E)	1.25
Electrical Conductivity (EC)	100			0.217
Nitrate Nitrogen (NO3-N)	94.9	105 (T3240022E)	103 (T3240022E)	2.54

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/28/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 2	T3240027B	Electrical Conductivity (EC)	EPA 120.1 and	0.226	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	145	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.8	95.2 (T3240023E)	96.4 (T3240023E)	1.25
Electrical Conductivity (EC)	100			0.217
Nitrate Nitrogen (NO3-N)	94.9	105 (T3240022E)	103 (T3240022E)	2.54

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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 10.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 1	T3240027C	Electrical Conductivity (EC)	EPA 120.1 and	2.22	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.69	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	1240	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	1.69	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1420	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	97.8	95.2 (T3240023E)	96.4 (T3240023E)			1.25
Electrical Conductivity (EC)	100			1.84 (T3240028D)	1.85 (Dup)	0.217
Nitrate Nitrogen (NO3-N)	94.9	105 (T3240022E)	103 (T3240022E)			2.54
Total Dissolved Solids (TDS)	104			905 (T3240033B)	910 (Dup)	0.551
Total Nitrogen (TN)	97.4			2.80 (T3240022B)	2.80 (null)	0.000

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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 2	T3240027D	Electrical Conductivity (EC)	EPA 120.1 and	2.27	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	1320	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	< 1.50	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1450	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	97.8	95.2 (T3240023E)	96.4 (T3240023E)			1.25
Electrical Conductivity (EC)	100			1.84 (T3240028D)	1.85 (Dup)	0.217
Nitrate Nitrogen (NO3-N)	94.9	105 (T3240022E)	103 (T3240022E)			2.54
Total Dissolved Solids (TDS)	104			905 (T3240033B)	910 (Dup)	0.551
Total Nitrogen (TN)	97.4			2.80 (T3240022B)	2.80 (null)	0.000

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PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 3	T3240027E	Electrical Conductivity (EC)	EPA 120.1 and	2.02	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	0.802	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	1120	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	< 1.50	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1290	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	97.8	95.2 (T3240023E)	96.4 (T3240023E)			1.25
Electrical Conductivity (EC)	100			1.84 (T3240028D)	1.85 (Dup)	0.217
Nitrate Nitrogen (NO3-N)	94.9	105 (T3240022E)	103 (T3240022E)			2.54
Total Dissolved Solids (TDS)	104			905 (T3240033B)	910 (Dup)	0.551
Total Nitrogen (TN)	97.4			2.80 (T3240022B)	2.80 (null)	0.000

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Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 4	T3240027F	Electrical Conductivity (EC)	EPA 120.1 and	2.27	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.233	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	1300	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	3.03	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1450	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	97.8	95.2 (T3240023E)	96.4 (T3240023E)			1.25
Electrical Conductivity (EC)	100			1.84 (T3240028D)	1.85 (Dup)	0.217
Nitrate Nitrogen (NO3-N)	99.9	101 (T3240027F)	99.6 (T3240027F)			1.65
Total Dissolved Solids (TDS)	104			905 (T3240033B)	910 (Dup)	0.551
Total Nitrogen (TN)	99.1			2.80 (T3240027F)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Live Oak Dairy F
PO Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/28/23

Tile Drain (H2)
SOURCE: Tile Drain

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Tile Drain	T3240027G	Electrical Conductivity (EC)	EPA 120.1 and	5.07	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	3270	mg/L
		Total Phosphorus (TP)	EPA 200.7	< 0.200	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	22.9	mg/L
		Soluble Salts (SALT-SOL)	Calculation	3240	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.8	95.2 (T3240023E)	96.4 (T3240023E)	1.25
Nitrate Nitrogen (NO3-N)	99.9	101 (T3240027F)	99.6 (T3240027F)	1.65
Total Dissolved Solids (TDS)	104			0.551
Total Phosphorus (TP)				

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Compliance Analysis Report

Customer
Live Oak Dairy F
PO Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/Ground Water (H6)
SOURCE: Canal Water

NOTES: Temperature was above qualifying at 13.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ponoche WD	T3240027H	Total Dissolved Solids (TDS)	SM 2540 C-1997 and	120	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	2.80	mg/L
		Electrical Conductivity (EC)	EPA 120.1 and	0.247	mmhos/cm

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Electrical Conductivity (EC)	100			1.84 (T3240028D)	1.85 (Dup)	0.217
Nitrate Nitrogen (NO3-N)	99.9	101 (T3240027F)	99.6 (T3240027F)			1.65
Total Dissolved Solids (TDS)	104			905 (T3240033B)	910 (Dup)	0.551
Total Kjeldahl Nitrogen (TKN)	99.1			2.80 (T3240027F)	2.80 (null)	0.000

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Live Oak F
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: FR
Sampled By: BN
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8/25	11:00	Domestic well	DW 1	11.6°		✓				
2	8/25	11:15	↓	DW 2	11.8°		✓				
3	8/25	11:30	Irrigation	IW 1	10.6°						H7
4	8/25	11:45	wells	IW 2	11.6°						H7
5	8/25	12:00		IW 3	12.0°						H7
6	8/25	12:15	↓	IW 4	12.0						H7
7	8/25	12:30	Tile Drain	Tile Drain		11.0					H2
8	8/25	12:45	Panoche WD	Canal		13.8°					H6
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	*(LABORATORY USE ONLY)* Logged in By:	LABORATORY NUMBER
1st						
2nd		DENELE ANALYTICAL, INC.	8-28-23 12:36			T3240027

Required to be submitted to any agency? (YES / NO)

LOF 2023 Annual Report

Final Audit Report 2024-06-22

Created:	2024-06-21
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAg-Jmu_PzjtSP0T_8tHY7ryxhLghXrwec

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